Timed RIS Control in the Loop: A Sample-Accurate IQ-Level Emulator with Characterized Timing

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***Abstract*—Controllers of reconfigurable intelligent surfaces (RIS) are usually tested on per-block channel coefficients, blind to when a switch lands in the signal. We present sremu, a software-only emulator that applies RIS-assisted propagation to complex-baseband (IQ) samples of external applications and executes timestamped commands of an in-the-loop controller at an exact, logged sample on every receiver. In lockstep runs with modelled latency and a blocked line of sight, the IQ-level packet error rate follows the closed-form probability that a switch lands after the preamble; unblocked, 54–99 % of such packets fail at $E_s/N_0 = 15$ dB. On an AWS c7i.8xlarge (Linux), sremu alone (headless) sustains 8 transmitter–receiver links at 1 MS/s, missing under 10^{-3} of 1 ms page deadlines; with 2 links serving an external controller's 10–100 commands/s, it applied at least 99.8 % of commands timed 4 ms ahead at their exact sample, but missed 7.7–26.7 % of these deadlines. With an offline coefficient evaluator, position-driven binary control loses half its benefit after 0.27–0.55 λ/v of signaling delay (wavelength over speed) over 12 geometries. The emulator reproduces a published path-loss model, inheriting its 0.98 dB mean offset from third-party 27 GHz chamber measurements; no switching or timing behaviour was compared with measurements.**

***Index Terms*—Reconfigurable intelligent surface, smart radio environment, IQ-level emulation, software-defined radio, controller in the loop, signaling delay, channel emulation, verification and validation.**

I. Introduction

RECONFIGURABLE intelligent surfaces (RIS) are planar arrays of passive elements whose reflection states are set by a controller, so that part of the propagation environment becomes a configurable system component [1]–[3]. Standards bodies now describe their use cases, architectures, channel models and test methods [4]–[7]. Much of the literature asks how a controller should choose a configuration: with discrete phase states [8]–[10], phase-dependent amplitudes [11], receiver mobility [12], [13] or sensing [14].

Most of these controllers are evaluated with ideal element states, instantaneous reconfiguration and one channel coefficient per block. Hardware departs from each: unit cells have state- and angle-dependent responses [35], [36] with coupled amplitude and phase [11]; the specification GS RIS 008 of the European Telecommunications Standards Institute (ETSI) lists reconfiguration time and modulated-signal quality among the parameters of a RIS test [7]; and a command travels over a control link with its own delay, so a position-driven controller acts on a stale position [13]. A reconfiguration inside a packet shows in the received samples, not in a per-block coefficient.

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Existing tools cover parts of this problem (Section II, Table I): coefficient-level simulators run offline [15], [17], Open Radio Access Network (O-RAN) frameworks loop over coefficients [26], [27], and IQ-level emulators have no RIS model [31]–[33] or fix the RIS channels in advance [28], [29]. To our knowledge, as of 10 Oct 2026, no surveyed tool combines (i) multi-node emulation at the level of complex-baseband (IQ) samples behind an interface in the style of a software-defined radio (SDR); (ii) RIS configurations issued at run time by an external controller and applied at an explicit, logged sample; (iii) per-state responses imported from measurements as functions of state, frequency, and incidence and reflection angle in one plane; and (iv) characterized wall-clock behaviour. Each property exists in some tool; the contribution of this paper is their integration, together with the evidence needed to trust the result.

We present `sremu`, a software-only emulator of a RIS-assisted radio environment at the level of complex-baseband samples, to which external transmitter, receiver and controller processes connect through an SDR-style interface. Time is an exact rational sample count. A RIS command requested for the switch time t_{sw} at the surface takes effect on receiver r at the sample nearest to $(t_{\mathrm{sw}}+\tau_r)f_s$, where τ_r is the propagation delay from the surface to that receiver and f_s its sample rate, and both the requested and the applied sample are logged. The paper makes four contributions.

- *Timed RIS control in an IQ-level emulator* (Section III). The switch time is defined at the surface and applied per receiver, a late command keeps one physical switch time for all receivers, and every command is logged; in lockstep mode, virtual-time barriers make controller-in-the-loop runs deterministic, in process and over the control socket, and a paced mode measures its own timing. The interfaces are a Python application programming interface (API) shaped after the USRP Hardware Driver (UHD) [37], a framed ZeroMQ (ZMQ) data plane, and a response-table schema aligned with the parameters of ETSI GS RIS 008 [7].
- *A verified physical model with stated validity limits* (Sections IV, V). An area-tied aperture kernel with a cell model and a model card; an import path for per-state tables, tested but not used here with a measured table; 58 analytic oracle checks and MATLAB code-to-code checks; agreement of the model class with third-

TABLE I
SURVEYED TOOLS AGAINST THE FOUR PROPERTIES OF SECTION I (AS OF 10 OCT 2026): (I) LEVEL AND INTERFACE, (II) RUN-TIME RIS CONTROL, (III) RIS MODEL AND MEASURED PER-STATE RESPONSES, (IV) WALL-CLOCK BEHAVIOUR. NV: NOT VERIFIED FROM A PRIMARY SOURCE; NO EVID.: A PRIMARY SOURCE DOES NOT MENTION IT. COEF.: CHANNEL COEFFICIENTS; CIR: CHANNEL IMPULSE RESPONSE; TS: TIME SERIES; IQ: COMPLEX-BASEBAND SAMPLES; DES: DISCRETE-EVENT SIMULATION; PO: PHYSICAL OPTICS. APPENDIX A GIVES VERSIONS, DATES, MOBILITY AND LICENCES.

Tool	(i) Level	(ii) Run-time RIS control	(iii) RIS model; measured responses	(iv) Wall-clock	Open
SimRIS [15], [16]	Coef.	No	Statistical, ideal phase; no evid.	No	Yes
QRIS [17], [18]	TS (QuaDRiGa)	no evid.	Unit-cell patterns; yes	No	On request
Sionna RT [19], [20]	Coef./CIR	Gradient opt. (v0.18–0.19)	Reradiation until v0.19; partial	No	Yes
MATLAB [21], [22]	Coef., samples	User script	Element-pattern cell; partial	No (offline)	No
Wireless InSite EES [23], [24]	Ray paths	No	Static PO metasurface; nv	No	No
CoopeRIS [25]	Packets (DES)	Oracle controller	Far-field phase sum; nv	No (DES)	Yes
GoSimRIS [26]	Coef., SNR	Yes (E2SM-RIS)	Configurable phases; no evid.	E2 (author claim)	Yes
ORIX [27]	Coef.	Yes (E2SM-RIS)	Quantized phase; parametric fit	Near-RT loop	nv
Colosseum + RIS [28]–[30]	IQ (FPGA)	No (fixed in a run)	Precomputed cascade; no	Yes (hardware)	No
OAI rfsimulator [31]	IQ (TCP)	Telnet, no RIS	None; no	No (by design)	Yes
srsRAN 4G ZMQ [32]	IQ (ZeroMQ)	User-built	None; no	nv	Yes
ACHEM [33]	IQ, UHD API	Channel, no RIS	None; antenna patterns	Yes	Yes
TRACE [34]	Symbols	Yes (UDP)	Ideal phase; no	nv	On acceptance
`sremu` (this work)	IQ, UHD-shaped API	Yes, at a logged sample	Area-tied kernel; one-plane $\Gamma(s, f, \theta_i, \theta_s)$ tables; import tested, not with measured data	Lockstep exact; paced measured	Yes (MIT)

party chamber measurements, inherited from a published path-loss model that the emulator reproduces [36]; a consistency check against a laboratory measurement from the author's earlier group work under a pre-specified plan hash-locked before unblinding, with its null result and measurement-design lesson; and carrier-frequency-offset (CFO) and bit-error-rate checks against the Cramér–Rao bound and theory.

- *Characterized wall-clock behaviour* (Section VI). Definitions of lockstep, paced, headless and served operation, the real-time factor (RTF) and deadline misses; scaling over links, elements, surfaces, threads, sample rates and feature sets, including the native ZMQ data plane, on AWS Linux servers; and, for a controller served over the control socket, command latency and timed-command exactness, though page deadlines fail our Sustainable rule.
- *Case studies of timed control* (Section VII). At the sample level, in lockstep with an in-process controller and a modelled command latency: what a reconfiguration inside a packet does to the symbols, and when timestamped commands, commands applied on arrival and switching on a free-running timer lose packets. On channel coefficients (for the mobility study E-A, an offline evaluator that agrees with sample-level IQ runs within the pre-specified tolerance, check VA-1): the time scale on which signaling delay erodes position-driven binary control in a blocked corridor, and a non-ideality ablation.

Scope. `sremu` is not a UHD drop-in: it offers a UHD-shaped API, and virtual-USRP emulation is a separate line of work [33]. It is not a hard-real-time emulator, not a ray tracer, and not a model of mutual coupling beyond what per-state tables measured with identical neighbours contain [6]. Fields are scalar and co-polar, RIS paths are single-bounce, and wall reflections are of order one. Paced-mode timing is measured and reported per platform, never assumed. Appendices A–F, after the references, extend Sections II–VII; Appendices G–I describe the oracles, the build and the provenance of every result.

II. RELATED WORK AND GAP

A. Coefficient-level simulators and ray tracers

SimRIS generates statistical millimeter-wave RIS channels with ideal phase control [15], [38]; QRIS extends QuaDRiGa [39] with measured or full-wave unit-cell patterns and several surfaces, with code on request [17], [18]; NYUSIM-RIS extends a statistical channel model [40], and system-level simulators [41], [42] abstract the links. Among ray tracers, Sionna added RIS in version 0.18 [19], and Sionna RT [43], [44] removed it in version 1.0; no release up to 2.2.0 (29 Sep 2026) has a RIS module [20]. Wireless InSite models engineered surfaces in a static configuration [23], [24], other ray tracers model RIS reradiation with ray-based or macroscopic models [45]–[49], and MATLAB provides RIS objects with a user-defined unit cell that process sample vectors offline [21], [22]. Electromagnetic models with mutual coupling [50], [51] and macroscopic reradiation models [52] describe the surface rather than a running link. Here the RIS configuration is an input to an offline computation, not a command received while a link runs; a systematic review co-authored by the author covers this group in more breadth [53].

B. Network simulators and controller-in-the-loop frameworks

CoopeRIS couples a far-field RIS gain model to vehicular network simulation with an oracle controller that knows the vehicle position [25]; ns-3 modules model links assisted by intelligent reflecting surfaces at packet level [54], [55] or draw channels from Sionna RT [56]. GoSimRIS closes an O-RAN E2 loop between an xApp and a RIS channel simulator through an E2SM-RIS service model and supports at most 64 elements [26]. ORIX runs a near-RT loop of 10 ms to 1 s with quantized phases and a phase-dependent amplitude model, and neglects mutual coupling and reconfiguration delay [27]. These frameworks exchange coefficients, signal-to-noise ratios or rates, not waveform samples; a hardware demonstration closes the loop with a physical surface [57].

C. IQ-level emulators

Colosseum combines 256 software-defined radios with a channel emulator on field-programmable gate arrays that runs precomputed scenarios [30], [58]. Tsampazi and Melodia installed RIS-cascaded QuaDRiGa channels in it and ran srsRAN end to end, with RIS phases optimized before the experiment and not reconfigured during it [28], [29]. The OAI rfsimulator forwards IQ samples over TCP, is by its documentation not designed to keep pace with wall-clock time, and changes channel models and per-beam gains during a run through a telnet interface [31]; the srsRAN ZMQ virtual radios exchange IQ samples, and a GNU Radio flowgraph can insert channels [32]; software-in-the-loop emulators add channel models to such transports (Appendix A). ACHEM runs unmodified UHD applications through a virtual USRP, emulates channels at the IQ level at wall-clock rate with mobility from vehicle simulators, and reports end-to-end validation against the AERPAW testbed; it has no RIS model [33].

D. Open RIS hardware and measurement data

Open RIS designs publish hardware, firmware and measurements [59]–[61]. Rossanese et al. released sub-6 GHz chamber beam scans [62] of a surface with seven reflective phase states and an absorbing state [63], the next validation step for `sremu` (Section IX); this paper compares against the chamber measurements of Tang et al. [36] (Section V-C).

E. TRACE

TRACE [34], by colleagues of the author, is a socket-based modular framework that decouples transmitter, radio environment, controller and receiver through separate control-plane and data-plane interfaces. It evaluates minimum mean-square-error (MMSE) and orthogonal-matching-pursuit (OMP) channel estimation and a differential channel-aware update (DCAR) on symbol-level i.i.d. Rayleigh channels with Gaussian random-walk and Gauss–Markov evolution, with an ideal continuous-phase, unit-amplitude surface, and reports a control-plane round-trip time of 5.2 ms over UDP [34]. TRACE and `sremu` are complementary: TRACE studies estimation and tracking algorithms on symbol-level channels; `sremu` provides a sample-level radio environment with timed RIS control. A TRACE-style controller could drive `sremu`'s control plane.

F. Gap

We surveyed RIS simulators, ray tracers, network simulators, controller frameworks, IQ-level emulators and open RIS hardware and data releases up to 10 Oct 2026 (method in Appendix A). Table I shows that each of properties (i)–(iv) of Section I appears in some row and no surveyed tool has all four. ACHEM has (i) at a stronger level than `sremu`, since it runs unmodified UHD applications, and (iv), but no RIS [33]; GoSimRIS and ORIX have (ii) at the level of channel coefficients [26], [27]; QRIS accepts measured unit-cell patterns, close to (iii), but runs offline [17]. In `sremu`, (iii) is a tested import path that this paper exercises with reported and assumed states, not with a measured per-state table (Section V). The gap is therefore a combination, and it is dated: a RIS operator could be added to an existing IQ-level emulator (Section IX).

An earlier simulator written by the author was used in [13], [14]; `sremu` is an independent new implementation. The laboratory data of Section V-D, the frequency-offset sweep of Section V-E and the FM scenario appeared in an unpublished ANTS 2025 submission by the author and colleagues, whose qualitative validation Section V-D supersedes.

III. Design

The tests that check statements of this section and the next are listed in Appendix G, each with the subsections whose statements it checks.

A. Architecture

The engine computes the samples of every receive (RX) stream from the samples of every transmit (TX) stream, the propagation model and the RIS configurations in force (Appendix B draws the processes and planes). It runs inside the Python process of an application (`Device.inproc`) or as a served process reached through sockets (`Device.connect`); both forms implement one engine interface, and a run whose TX and RX applications are separate operating-system processes gives the same RX samples, bit for bit, as the same run in one process. A served engine uses ZMQ sockets [64]: a framed IQ data plane (one ingress socket for TX applications, one egress socket per RX stream), a raw request–reply data plane (Section III-C), a control plane of JavaScript Object Notation (JSON) requests on one `ROUTER` socket (by default `tcp://127.0.0.1:5600`), which also carries the barrier messages of registered controllers, and a `PUB` socket for telemetry. A recorder writes a manifest, the resolved scenario, an event log of every command record and, optionally, SigMF recordings [65] per run. `sremu` is a Python package (Python 3.12 or later; 103 modules, about 41 thousand lines) with a NumPy reference backend and a compiled numba backend for rendering (dependencies in Appendix B); it runs on Windows and Linux.

B. Time model and command semantics

Sample rates and virtual times are exact rational numbers, and sample 0 of every stream is at $t = 0$. A time maps to a sample as $\mathrm{to_ticks}(t, f) = \lfloor tf + 1/2 \rfloor$, the nearest sample with ties rounded up as in UHD [37], and a stream produces $\lceil t_1 f \rceil - \lceil t_0 f \rceil$ samples on $[t_0, t_1)$ for any mix of rates; wall-clock time never enters data-plane arithmetic. Each RX stream is processed in pages of about 0.5 ms on an absolute grid, cut at snapshot boundaries, switch samples, barrier limits and the end of TX input (Fig. 1(a)). Per-sample quantities are closed-form functions of the absolute sample index, and sums run in a fixed order, so the RX samples are identical bit for bit however clients chunk their calls, wherever events cut a page and, in deterministic mode (the default, Appendix B), whatever the thread count.

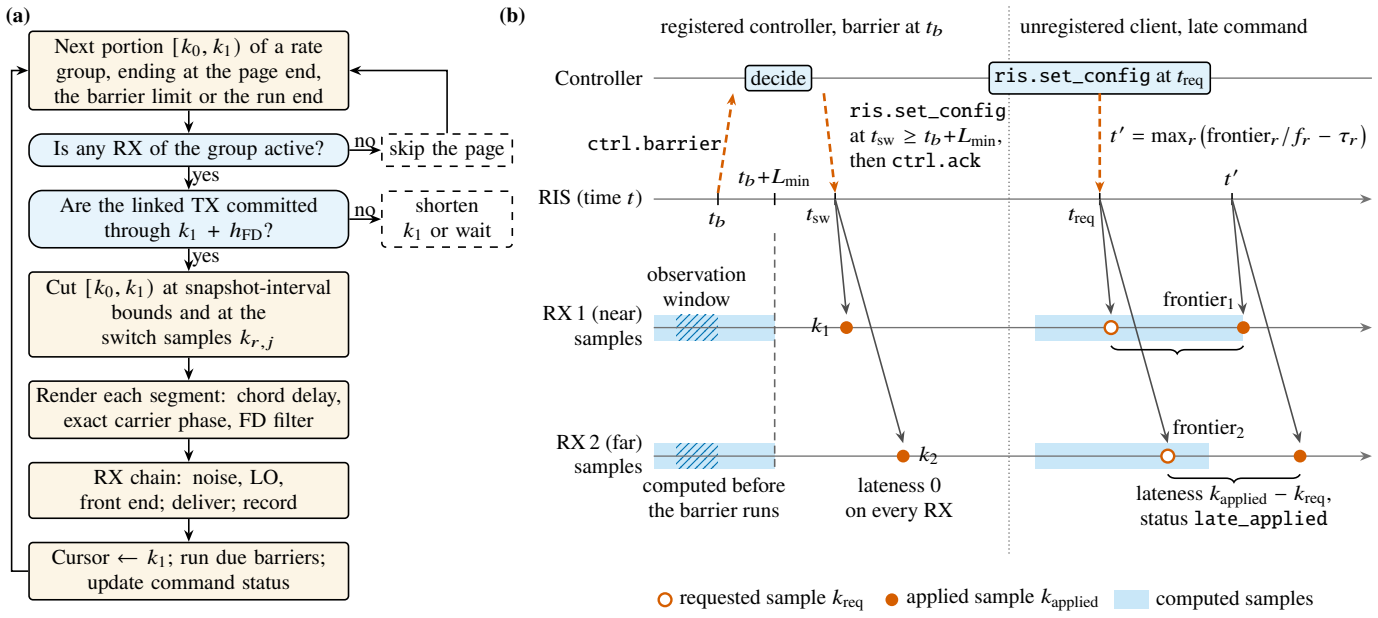


Fig. 1. (a) Processing of one page portion of a rate group; h_{FD}: fractional-delay filter look-ahead. (b) Life cycle of RIS commands on the time axes of the surface and of two receivers. Left: a registered controller decides at barrier t_b after every receiver has been computed up to $t_b + L_{min}$, so its command at $t_{sw} \geq t_b + L_{min}$ lands without lateness at $k_r = \text{to_ticks}(t_{sw} + \tau_r, f_r)$. Right: a command that arrives after its samples were computed is moved to the single surface time t' that every receiver can still honour; the record keeps requested and applied samples.

RIS commands: A RIS command carries a switch time t_{sw} defined at the surface. Receiver r sees the change on the paths of render tile j at sample

$$k_{r,j} = \text{to_ticks}\big(t_{sw} + \tau_{r,j}(t_{sw}),\ f_r\big), \tag{1}$$

where $\tau_{r,j}$ is the propagation delay from the reference point of tile j (the panel centre for an untiled panel) to the receiver in the geometry at t_{sw}, so receivers at different distances see one switch at different samples (Fig. 1(b)). The time of a command is an exact surface time, a sample of a named RX stream, or 'as soon as possible' (asap), the earliest time that every receiver can still honour. The frontier of a receiver is its earliest uncomputed sample, and a command is late when a switch sample falls below a frontier. The default policy, `apply_on_arrival` as in UHD, moves the switch to a single surface time t'. It starts from $t' = \max_{r,j}(\text{frontier}_r / f_r - \tau_{r,j})$, with the delays taken at the requested time (at the latest frontier for asap), maps every receiver again with (1), which takes the delays at t', and raises t' until every $k_{r,j}$ reaches frontier_r, so under mobility no switch lands in the past and all receivers still see one physical switch. The binding receiver's switch arrives at its frontier instant rather than up to half a sample earlier, so a receiver with a higher rate may switch a few samples later than rounding alone would allow; the policy `reject` returns `LATE_COMMAND` and applies nothing. Each command yields one record with requested and applied surface times and samples per receiver, their difference (the lateness), wall-clock receipt and delivery times, status and a hash of the applied configuration. A reconfiguration need not be instantaneous: changing elements form transition classes with profiles g_c (a step, the default; a linear ramp; or an exponential), and because combine is linear in Γ the channel is exactly $h(t) = h_{old}(t) + \sum_c g_c(t - t_{sw,r})\,\Delta h_c(t)$, with each change Δh_c interpolated like any path, also under mobility. The default duration is the panel's `reconfiguration_time_s`, named after a GS RIS 008 parameter [7]; the complex-linear path between states is an engineering choice (ENG).

Controllers in the loop: A controller registers with a period T_c and a minimum latency L_{min}. At each barrier $t_b = mT_c$ the engine computes every receiver up to sample $\text{to_ticks}(t_b + L_{min}, f_r)$, exclusive, and then gives the controller an observation built from samples before $\text{to_ticks}(t_b, f_r)$, such as each receiver's measured power. Actions take effect at t_b plus a latency drawn from a keyed model and never smaller than L_{min}; an earlier action raises `LOOKAHEAD_VIOLATION`. The frontier limit and (1) use the same monotone rounding and $\tau_{r,j} \geq 0$, so these actions are never late and the time a controller takes to decide changes no RX sample, in process or remote: the lookahead rule of conservative parallel discrete-event simulation.

Lockstep, paced and batch operation: In lockstep mode virtual time advances as far as the inputs allow: a TX stream is committed through sample K when its samples cover K, its last burst ended, or it announced idleness up to K (`idle_until`, an extension of the UHD interface). Runs with barrier controllers are deterministic; other timed commands are deterministic when they arrive on time, and otherwise the manifest flags nondeterministic inputs. In paced mode, page j is released at wall time $R_j = t_0 + k_1/f + D_{lag}$, with page end k_1 and an input slack D_{lag}; late inputs are dropped, missing samples become zeros and a slow RX client loses pages, each with a report, and deadline misses are counted. Paced mode is soft real time: it counts misses rather than assuming their absence, and it admits asynchronous con-

trollers, whose commands carry measured lateness. A batch path (`run_batch`) evaluates $\mathrm{combine}(\mathrm{snapshot}(t), \mathrm{cfg}(t))$ on a time grid without IQ samples, with the same model, command semantics (switches at their surface time), controllers and random keys; in a static scene with noise off, an in-process controller and no signaling delay, its time-averaged received power equals that of the IQ path within a relative 10^{-9} (VA-4). Holding each step at its start value, it reads up to 0.09 dB high under mobility with a delayed controller and fails the limit of VA-1, so E-A uses a midpoint evaluator (Section VII-B). A replay re-injects the recorded TX streams and commands at their applied times and reproduces the recorded RX samples bit for bit.

C. Interfaces

The Python API follows UHD [37] without binary compatibility: streams carry UHD-style metadata with exact time specifications, failures are status codes, and RIS commands carry state indices, phases or a codebook index, an optional transition and a late policy. Each stream has one channel, and an unmodified UHD application does not run against this interface (Section IX). Control messages are validated before dispatch, so every refusal has one code, and a socket command lands at its exact sample with zero lateness in lockstep mode. Sockets bind to loopback addresses unless remote binding is allowed; there is no authentication. For radio software that exchanges raw samples over request–reply sockets, the engine takes the broker role of the srsRAN ZMQ setup [32]: it requests TX samples from the peer and answers its receive requests, with timing implicit in sample counts, so the mode is lockstep. Its two request formats, taken from the srsRAN 4G and GNU Radio sources, are tested against stub peers, and unmodified GNU Radio 3.10.9.2 runs against it (Section VII-D); no srsRAN binary has been run, so we call it an srsRAN-style raw-IQ mode.

D. Element response tables

Element responses are tables $\Gamma(s, f, \theta_i, \theta_s)$ over state, frequency, and signed incidence and reflection angles in one plane of incidence (Appendix C), read at the carrier, with mandatory provenance such as the kind of source, uncertainties and a hash of the raw data. Validators reject tables that are not passive, not reciprocal or too coarse, and queries outside the grid or band; interpolation is complex-linear, so the channel stays linear in Γ. Tables come from CSV files with a provenance sidecar, MATLAB unit-cell curves or built-in alphabets, including the published states of Tang et al. [36]. Panel capabilities carry the names of GS RIS 008 parameters [7] where they exist (Appendices A, B).

IV. Physical Model

A. Scene and snapshots

Node positions follow analytic trajectories in continuous time. At a snapshot time the model evaluates, in double precision, every path without a RIS and, per panel, a coefficient tensor $B[r, s, n]$ over rows r (a link with a ground variant of its two hops), states s and cells n; a configuration selects one state per cell, and $\mathrm{combine}$ forms one path per panel, row and render tile, linearly in Γ. The line-of-sight (LOS) path follows the Friis equation with scalar co-polar antenna patterns and an explicit visibility factor (an occluder, a partition loss or a coherent knife edge), which never acts on a RIS hop; ground, order-one walls and a diffuse term are optional (Appendix C).

B. RIS kernel K1

The RIS contribution of every link is the area-tied aperture kernel K1

$$\begin{aligned} h_{\mathrm{RIS}} = \sum_q & \sqrt{G_T(\hat{r}_q^i)\, G_R(-\hat{r}_q^s)}\; \Gamma_{n(q)}(s, f, \theta_{iq}, \theta_{sq}) \\ & \times \frac{A_q\, o(\theta_{iq}, \theta_{sq})}{4\pi\, r_{iq}\, r_{sq}}\; j\, e^{-jk(r_{iq}+r_{sq})} \\ & \times \mathbf{1}[\cos\theta_{iq} > 0,\ \cos\theta_{sq} > 0], \end{aligned} \tag{2}$$

where q runs over quadrature points (one per physical cell, or sub-cells, Section IV-C) and $n(q)$ is the cell that carries the state and Γ; A_q is the pitch area of the quadrature cell, gaps included, so the sub-cell areas of a cell add up to its pitch area; r_{iq}, r_{sq} are the distances from transmitter and receiver to the point and θ_{iq}, θ_{sq} the angles toward them from the panel normal (projected onto one plane for Γ, Section III-D); $\hat{r}_q^i$, $\hat{r}_q^s$ are the unit vectors from transmitter to point and from point to receiver; $o = \sqrt{\cos\theta_i \cos\theta_s}$ is the default obliquity, the cosine pattern measured by Tang et al. [36]; j is the Huygens (physical-optics) factor; and the indicator keeps front-side incidence and reflection, so a path to the back of a panel contributes exactly zero.

Because each cell's contribution is tied to its area, subdividing a fixed aperture into more cells leaves the received power unchanged (Section V-C), whereas a fixed per-cell gain would grow with every subdivision; and the factor j sets the phase of the RIS path relative to the LOS path, which would otherwise be a quarter turn off (image theory). For one quadrature point per cell, K1 is algebraically equal to three published forms: the cell model of Tang et al. [36] times j, the general path-loss model of ETSI GR RIS 003 [6, clause 6.3.1.2] with its recommended gain $4\pi d_x d_y/\lambda^2$ of a $d_x \times d_y$ cell, and the model of Ellingson [66] with an element gain tied to the cell area, $G_e(\psi) = (4\pi A/\lambda^2)\cos\psi$, to double precision.

C. Cell model

Each panel selects a cell model: `point`, one quadrature point per cell, which Tang et al. compared with their measurements [36]; `po_patch`, sub-cells with sides of at most 0.2λ that share the cell's Γ, a uniform-patch model; or `auto` (default): `point` for pitches up to 0.5λ, else `po_patch`. A patch carries a sub-aperture factor that a point omits: for a half-wavelength cell -3.9 dB and -8.9 dB at 30° and 45° retro-reflection, none at specular reflection. Every case study uses `point`, though the laboratory tiles of 42.8 mm lie 0.03 mm below the threshold; forcing `po_patch` on them in E-B and E-C moves no loss or switch effect by more than 1.5 dB (Appendix C).

D. Element states and non-idealities

Each cell's reflection coefficient comes from its response table through a fixed pipeline: expansion by the panel's control granularity, quantization of continuous phases by $s = \arg\max_s \mathrm{Re}\{\bar{\Gamma}(s)e^{-j\phi}\}$, with $\bar{\Gamma}$ the table at the panel's reference geometry, correct for unequal amplitudes, a failure map, the table at the cell's angles, an efficiency factor, and a coupling hook, implemented as no coupling. Mutual coupling is not modelled beyond the locally periodic coupling that a per-state table taken with identical neighbours contains [6]. Closed-form losses of finite alphabets and the mirror and steerable parts m, d of a binary alphabet $\Gamma = m \pm d$ check the pipeline. An exact maximizer of $|z_0 + \sum_n Z[s_n, n]|$ over any finite alphabet, a sweep over sorted critical angles (Appendix C), bounds controllers and studies and matches exhaustive search in every tested instance; optimal discrete-phase beamforming has been solved before [9], [10].

E. Mobility, delay and noise

Doppler shifts are not parameters: they follow from the geometry. Snapshots lie on an absolute grid of period T_u; between two of them each path without a RIS has a delay linear in time and the exact carrier phase $e^{-j2\pi f_c \tau_p(t)}$ for that delay, and each RIS panel row is split into render tiles with a reference delay and an interpolated residual. Configuration switches are never interpolated across. One target, $\varepsilon_{\mathrm{rel}} = 10^{-3}$ for $|\Delta h|/|h|$, sets both the snapshot period and the tile size, and each run checks it at the midpoint of every 64th snapshot interval (Appendix C). Paths whose delay spread is too large for one envelope delay get Kaiser-windowed sinc fractional-delay (FD) filters [67] with an in-band error below -60 dB. Thermal noise of power $k_B T_0 F f_s$, with $T_0 = 290$ K and the receiver noise factor F ($k_B T_0$ is -173.975 dBm/Hz), is added once per receiver, not per link. Every random variate is a keyed pure function of the seed and an absolute sample or step index, so adding a node or chunking a run changes no other variate. Oscillator offsets, phase noise and a front end with optional clipping complete the impairments; Appendix C gives the model card with the validity domain, evidence and default of every component.

V. Verification and Validation

Verification asks whether the code computes what the model states; closed forms, exact properties and an independent code answer it. *Validation* asks whether the model describes the physical world, which needs measurements; agreement with MATLAB is therefore verification. Data read off a published figure are labelled DIG with their digitization uncertainty, and every caption names the classes of its data (Appendix I).

A. Analytic oracles

The test record of emulator commit 1e1fcab is summarized per tier in Appendix D. Each identifier (Appendix G) names one property, its oracle (a closed form or exact property, kept in a test package that may not import the code under test) and a fixed tolerance. All 58 analytic identifiers (PH, RT and IO) pass, in 674 implementation tests; the statistical tier adds 5 identifiers and the code-to-code tier 8. In total 2136 tests passed and 5 were skipped, none of them with an identifier.

B. MATLAB code-to-code comparison

Eight golden references, G1–G8, from MATLAB R2026a scripts cover the per-element kernel, beam patterns, frequency offset, thermal noise, fading statistics, the far-field closed form of ETSI GR RIS 003 [6] and a two-ray channel; all 8 pass. They show that two codes agree; they are not validation (Appendix D).

C. Published RIS path-loss model and measurements

Fig. 2(a) checks a property that published curves cannot test: when a fixed aperture is cut into more cells, the received power must not change. Far from the panel, K1 with the default cell model stays within 0.006 dB of the closed form for every cell count from 16 to 16 384 (PH-04); near the panel, the patch model stays within 0.04 dB of the continuous physical-optics integral, while one point per cell converges slowly (Appendix D).

Tang et al. measured the path loss of millimeter-wave surfaces in an anechoic chamber and fitted a model to it [36]. We extracted the curves and points of their Fig. 10 [36] (DIG; Appendix D-C) and ran the same 36 links through the emulator with the default cell model and the reported unit-cell state and link budget (REPORTED). The emulator reproduces the published model curves within 0.02 dB; since K1 equals that model for one point per cell, this checks the implementation and bounds the digitization error. Against the chamber measurements the emulator's received power is low by 0.98 dB on average (SD 0.38 dB, RMSE 1.05 dB; per geometry, with the caveats of the digitized data, in Appendix D-C), every measured point lies above it (Fig. 2(b)), and the published model is low by 0.98 dB: the emulator inherits the published residual. This is agreement of the model class with chamber data for one 27 GHz surface in one uniformly coded state in specular geometries; it tests the absolute level and its dependence on distance and angle, not reconfiguration, steering or near-field structure, and no switching, transient or timing behaviour of the emulator has been compared with measurements.

D. Laboratory consistency check

The data set is a laboratory measurement collected at IIT Bhilai for that earlier submission (Appendix D): the received power at 3.5 GHz on a polar grid of 30 cells in front of a binary RIS of six tiles, whose cell states have phases of 49.64° and $-153.77°$ (magnitudes unknown), surviving as a colour map that we digitized (DIG, 0.045–0.22 dB per cell). Its geometry is known within ranges, the measurement positions to about ± 2 cm.

We therefore wrote the analysis plan before fitting any RIS model to the data: 2000 geometries drawn from a prior over the unknowns, five models with fitted offsets, among them line of sight alone (M0), a log-distance fit (M0n) and the laboratory configuration (M2), fitted on three rings and scored

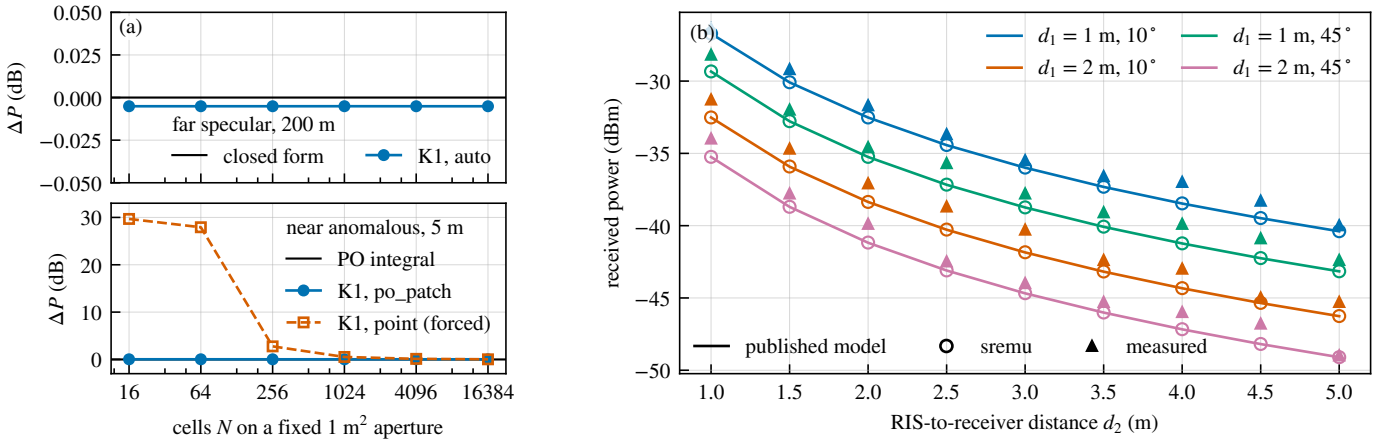


Fig. 2. (a) Fixed-aperture refinement of K1: deviation of the received power from a reference as one aperture is cut into N cells. Top: far specular reflection, default cell model, against the far-field closed form. Bottom: near-field anomalous reflection, patch model and a forced point model, against the continuous physical-optics integral. (b) Published model curves and digitized chamber measurements of Tang et al. [36] and the emulator's values for the same links, four geometries. Data classes: ANALYTIC (references in (a)), SIM-sremu (emulator), DIG (published curves and measurements), REPORTED (link budget).

on two, and three decision rules: C1 (level consistent), C2 (the RIS model explains spatial structure better than M0) and C3 (better than a uniform mirror). The SHA-256 hash of the analysis plan, which the data loader checks, was pushed to the repository after a descriptive inspection of the map but before any RIS model was fitted to it (Appendix D): the plan is pre-specified and hash-locked before unblinding, not blind, and the lock rests on the repository's push log, not on an external registry. In a power analysis on 500 synthetic maps drawn from M2 itself, the full analysis found C2 in 0 of them (upper limit of the two-sided 95 % Clopper–Pearson interval 0.7 %), far below the pre-set threshold of 0.5, so the study was labelled underpowered before the data were opened.

The outcome is a null result: the median test RMSE is 3.96 dB for M0, 3.77 dB for M0n and 5.11 dB for M2, the RMSE gain of M2 over M0 is -1.15 dB with a Spearman correlation of -0.31, and C1, C2 and C3 all fail, also in every pre-set sensitivity run. Since the comparison is underpowered, this is not evidence for or against the model. A diagnostic outside the protocol found that no single extra fact lifts the C2 detection rate above zero, while the exact as-built tile geometry with positions known to ± 1 mm gives 0.53. At 3.5 GHz the wavelength is 86 mm and the map's structure comes from path differences of a fraction of it, so a RIS map at this frequency needs its geometry recorded to about a millimetre before it can separate the models, and even then the detection rate stays near the pre-set threshold: a measurement-design lesson. The word *validation* is reserved for a data set with a recorded geometry.

E. CFO estimation and link metrics

With noise off, a receiver oscillator offset swept over 29 values within ± 1400 Hz is recovered by a least-squares phase estimator within 3.1×10^{-10} Hz, and the received samples are identical for three client epoch lengths and through two processes over the native ZMQ data plane (V-CFO-1, V-CFO-3). Exact rational sample counts avoid a generic pitfall: a time base that delivers a whole number of samples per tick scales every frequency estimate by the ratio of the two counts, an error of 6.9 % for ticks of 2 ms at 5880 S/s. With the emulator's thermal noise, a periodogram estimator reaches the Cramér–Rao bound of a tone in white Gaussian noise [68], within the pre-set limit of 1.5 times the bound at every gated offset (V-CFO-4, ST-05). Through a static lab-like link, the theoretical bit error rate (BER) of BPSK and QPSK with genie channel knowledge lies inside the 95 % Clopper–Pearson interval at all 12 points (study E-F, Appendix D).

F. Doppler and mobility

For a receiver passing a panel at 30 m/s and 3.5 GHz, the rendered line-of-sight path has the closed-form Doppler shift of 350.242 Hz within 2×10^{-10} Hz (PH-17), the RIS contribution follows an independent per-cell sum of K1 away from pattern nulls, and at the automatic snapshot period the largest relative interpolation error is 7×10^{-6}, below the target $\varepsilon_{\mathrm{rel}} = 10^{-3}$ (PH-18; Appendix D).

VI. Wall-Clock Behaviour and Scalability

Lockstep runs advance virtual time as fast as the inputs and the host allow and never drop a sample; paced runs release page j at the wall time R_j of Section III-B, so a received sample is delivered between D_{lag} (1 ms by default) plus the page's computation and one page more after its own wall time. The real-time factor (RTF) is the virtual time advanced (the minimum over the received streams) per unit of wall time. A paced page misses its deadline at slack s when it is delivered later than $R_j + s$; a configuration is *Sustainable* when, over at least 60 s after a warm-up of 5 s, fewer than a fraction 10^{-3} of its pages miss the deadline at 1 ms slack, no input sample arrives late and no output page overflows. A *headless* run generates and consumes its streams inside the emulator process, without client processes, transport or control socket, and bounds what the engine alone achieves; a *served* run exchanges samples or commands with other processes over sockets. A link is a transmitter–receiver pair ($L = n_{\mathrm{tx}} n_{\mathrm{rx}}$). The visible latency of a command runs from its receipt to the

delivery of the page that holds its switch; a timed command is late when that sample was already computed at its receipt, and is then applied on arrival. Lockstep benchmarks ran on AWS c7i.48xlarge instances, paced ones on c7i.8xlarge instances with the engine pinned to 16 logical CPUs, on scenes that load the engine and carry no physical result (Table X, Appendix E).

A. Lockstep scaling and data plane

One link at 1 MS/s through one surface of 240 elements, with a static geometry and thermal noise (feature set F1), runs at RTF 5.14; from $N = 40$ to 16 384 this RTF stays within 3 %, because a static snapshot is computed once, while with mobility (F3) it falls from 2.13 to 0.54. Surfaces likewise cost little when static and much with mobility (Fig. 10(d)): from none to four, the one-link RTF falls from 6.42 to 4.93 static and from 3.42 to 0.89 with mobility. With one static surface, 8 links keep RTF ≥ 1 at 1 MS/s for every N. Threads help little (Fig. 10(e)): at 8 links the RTF is 1.07 with one numba thread, 1.52 with 8 and 1.05 with 96. Up to 11 points ran at a time on one host; rerun alone, reference points were up to 16 % faster, so these values are conservative.

Served over the native ZMQ data plane to a transmitting and a receiving client process (F5, rerun B2-F45), the base point delivered samples end to end at RTF 0.90, against 1.42 in process with recording (F4) in the same run; over 32 configurations F5 reached 0.59–1.00 times the F4 RTF and never kept up with the wall clock (Fig. 10(f)), and all 100 stress runs of the base point on 10 concurrent workers completed. Appendix E adds the latency of the UHD-shaped API (E-F).

B. Paced operation

Headless (B3), at 1 MS/s with one surface of 240 elements and noise, the engine alone sustains 8 links with pages of 1024 μs (worst p99 page latency 848 μs), 4 with 512 μs and 2 with 256 μs; 8 links at 0.1 MS/s, 2 at 5.76 MS/s and one at 23.04 MS/s (Fig. 3(b)). These limits are indicative: link counts were doubled until a repetition failed, 6 of the 9 series ended on a single repetition, and short pages fail on computation or on rare stalls (Appendix E). Mobility, tried at 512 μs pages alone, missed 17 % of the deadlines with one link.

C. Control loop

In lockstep, controllers issued 22 800 commands at 10 Hz to 10 kHz, in process and over the control socket, and every lateness value is zero, as the barrier semantics require. From 100 Hz on, each in-process command costs 1.36–1.39 ms of engine time, so at 10 kHz the RTF drops from 2.53 to 0.071 in process and to 0.018 over the socket.

Served in paced mode to an external process sending configuration commands over the control socket (B4: 2 links, 1024 μs pages; Fig. 3(c, d)), the engine was not backlogged at 10 and 100 commands/s. Asap commands became visible after a median of 2.2–2.6 ms and a p99 of 5.1–6.0 ms; pooled over repetitions, timed commands took effect at their exact sample in at least 99.8 % of the cases with leads of 4 ms or more, and 3.3–5.7 % were late with 2 ms. No served repetition met the Sustainable rule: 7.7–8.2 % of the pages at 10 and 22.5–26.7 % at 100 commands/s missed the 1 ms deadline (at most 8.8×10^{-3} at 5 ms slack), where the same links and pages missed none headless. At 1000 commands/s the engine was the bottleneck (290–361 commands/s achieved). We thus regard served controller-in-the-loop operation, shown at 10–100 commands/s with leads of 4 ms or more, as preliminary.

Limitations. The numbers hold for the stated hosts, commits and scenes (one instance per benchmark, up to 3 repetitions per point). Paced runs, all on Linux, sustained at most 8 links at 0.1–1 MS/s and one at 23.04 MS/s headless; none exchanged samples with external applications. Paced mode is soft real time; its results are measured, not guaranteed.

VII. Case Studies

The studies use the semantics of Section III; their geometries are simulated (SIM) and labelled, and every assumed input is named. E-B, on when a command takes effect, runs at the sample level; E-A, on how old the information behind a command is, is evaluated offline on channel coefficients with a sample-level consistency check; E-C evaluates static channel coefficients; E-E runs unmodified radio software. Appendix F adds details and a model-based FM-band dimensioning (E-D).

A. Intra-frame RIS reconfiguration

A tool with one channel coefficient per block cannot place a reconfiguration inside a packet, yet ETSI GR RIS 004 sizes the delay budget of a RIS controller by the frame structure of 5G NR [69]. This study measures how a switch inside a packet and the timing of the command behind it interact, at the sample level and in lockstep with an in-process controller, so the command latency is modelled.

The geometry is modelled on the laboratory arrangement of Section V-D but simulated (SIM) with nominal dimensions (the middle of each known range); it is not the measured geometry, and no measurement is compared with these runs. On an arc of radius 1.95 m around the transmitter, six tiles of 5×8 cells act as one surface at 3.5 GHz; two receivers 1.2 m from the transmitter each have their own exact optimum, and a switch sends one command per tile for the same time. The cell states have unit amplitude (ASSUMED), transitions follow the complex-linear path (ENG), and the plate blocking the line of sight spares the RIS hops although it crosses 152 of the 240 transmitter-to-cell segments, an idealized regime (Appendix F). QPSK packets of 64 preamble and 512 payload symbols last 2.304 ms, the receiver equalizes once from the preamble, and $E_s/N_0 = 25$ dB at RX1.

In the mechanism experiment (E-B1) the surface switches to the configuration of RX2 during a packet to RX1. VA-5 checks every step switch inside a packet (16 on the arc, 16 in a lab-like geometry): the earliest affected sample of RX1 is exactly $\mathrm{to_ticks}(t_{\mathrm{sw}} + \tau_{\mathrm{RIS}\to\mathrm{RX1}})$, with the delay of each tile on the arc, and every earlier sample is identical bit for bit to the run without a switch. With the line of sight blocked (Fig. 4(a)), the switch changes the channel of RX1 by -23.7 dB and 137°. Symbols before it keep their error vector magnitude (EVM) of about -24.6 dB; after it the equalizer, trained on the old

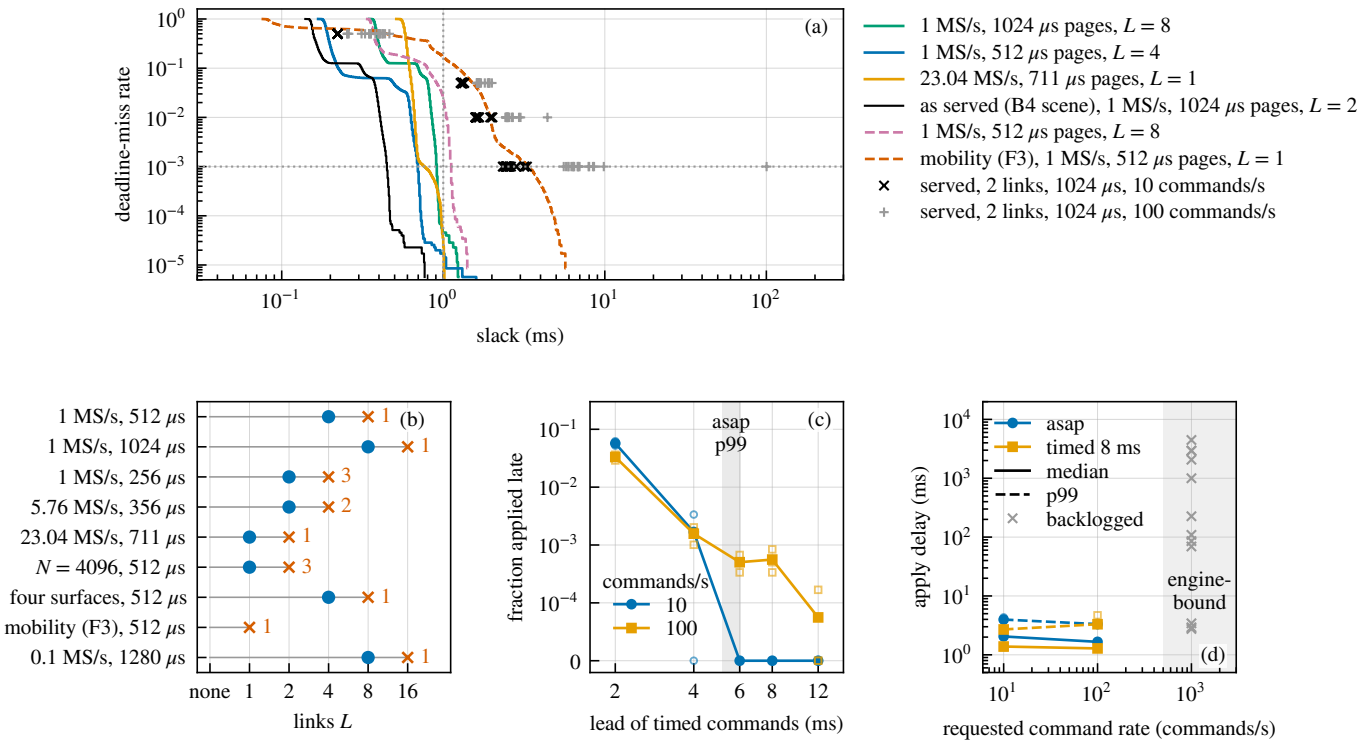


Fig. 3. Paced operation and the served control loop on AWS c7i.8xlarge instances (B3, B4; MEAS). (a) Deadline-miss rate versus slack (the complementary distribution of the page latency), pooled over the repetitions of headless points (solid: largest Sustainable points and the served scene; dashed: points that ended a series), and the reported percentiles of the served runs (2 links, 1024 μs pages; markers); dotted: the 1 ms slack and the 10^{-3} threshold. (b) Largest Sustainable number of links (dots; page duration as run; "none": not one link) and the failed count (crosses), labelled with the repetitions run there. (c) Fraction of served timed commands applied late versus lead (lines: pooled; markers: repetitions; zero at the floor); the others took effect at their exact sample. Shaded: range of the asap visible-latency p99. (d) Apply delay (after the nominal release of the page with the switch) versus requested command rate (median, p99); crosses: backlogged repetitions; shaded: engine-bound.

channel, sees an error as large as the signal, and 643 of 1024 payload bits fail when the switch comes a quarter into the packet. A linear transition of 200 μs spreads the damage over more symbols but still costs 577 bits. With the line of sight present, the switch changes the channel by -10.5 dB and 19°, the payload EVM rises to -3.5 dB, and the packet loses 1 of its 1024 payload bits: with a strong line of sight QPSK tolerates the wrong configuration.

In the timing experiment (E-B2) 2000 packets alternate between the two receivers, separated by guards of T_g. A controller sends one message per guard, which arrives after a modelled latency $\Lambda = \Lambda_0 + JU$, $\Lambda_0 = 0.2$ ms, U uniform on $[0, 1]$. Timed commands carry a switch time at the guard centre, are issued 0.5, 2 or 5 ms ahead and apply on arrival if late; commands without a time apply on arrival (asap), as untimed UHD commands do [37]. The whole grid was fixed before any point ran (Table XI; Fig. 4(b): $T_\mathrm{g} = 0.5$ ms). A packet is exposed when its switch lands after the preamble (T_pre), with probability $P(\Lambda > T_\mathrm{g} + T_\mathrm{pre})$ for asap commands and $P(\Lambda > \ell + T_\mathrm{g}/2 + T_\mathrm{pre})$ for timed ones with lead ℓ. With the line of sight blocked, the packet error rate (PER) follows this closed form (asap, $J = 1$ ms: 45.6 % against 44.4 %; timed, 2 ms lead, $J = 3$ ms: 23.1 % against 23.1 %), exceeding it by at most 8.8 points. The timing rules are thus designed properties of the semantics, which the emulator reproduces on IQ samples; the asap threshold moves with the guard (at $J = 1$ ms, 88 % lost with $T_\mathrm{g} = 0.1$ ms, none with 1 ms). The waveform adds the cost of exposure: a switch early in the preamble costs nothing, one late in it causes the excess, and with the line of sight present 54–99 % of the exposed packets fail at 15 dB and at most 3.1 % of all packets at 25 dB.

A third semantics, a constructed worst case, latches commands on a free-running 2 ms timer not aligned with the packets: its closed-form exposure (Appendix F) is 47–92 % without jitter over the guards (Fig. 4(b): 70–93 % lost), set by the timer period against the frame, not by the controller. RIS experiments with 5G NR testbeds also switch on an uncoordinated timer, but with a period of seconds [70], [71].

These runs are lockstep with a modelled latency; on the served path of Section VI-C, even the median asap latency exceeds every latency of the regime $J \leq 1$ ms of Fig. 4 (Appendix F). The lab-like geometry (Fig. 4(c), (d); Appendix F) supports this picture.

B. Signaling delay in closed-loop binary RIS control

A user walks a corridor of 10 m as a one-dimensional random waypoint at 1 m/s past a single row of on–off cells at 2 m (133 cells at 1 GHz, 320 at 2.4 GHz); a partition attenuates the line of sight by 20 dB but spares the RIS hops, although it crosses the path from the access point to every cell, an idealized regime. The exact optimum and the activation heuristic of [13] (rule in Appendix F) act on position reports

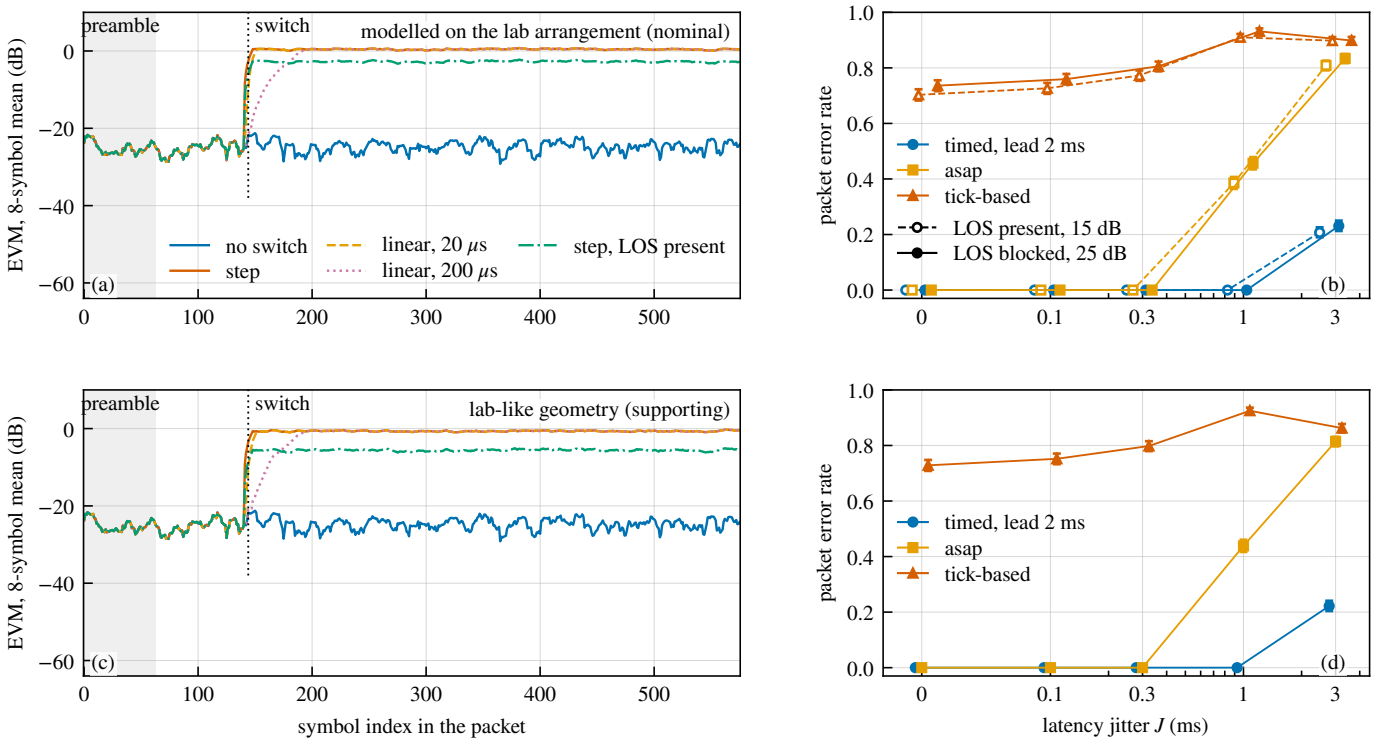


Fig. 4. Intra-frame RIS reconfiguration. Top row: geometry modelled on the laboratory arrangement (SIM, nominal geometry). Bottom row: lab-like SIM geometry (supporting), not the laboratory's. (a), (c) EVM along one packet, as a running mean over eight symbols, when the RIS switches a quarter into the packet: a step and two linear transitions with the line of sight blocked, a step with it present, and no switch. (b), (d) Packet error rate versus the jitter J (symmetric-log axis) of the command latency at a guard of 0.5 ms, for timed commands with a lead of 2 ms, asap commands and tick-based semantics (commands latched on a free-running 2 ms timer), with Clopper–Pearson 95 % intervals. Filled markers: line of sight blocked, $E_s/N_0 = 25$ dB. Open markers, in (b): line of sight present, $E_s/N_0 = 15$ dB. Data classes: SIM-sremu (emulator runs), ASSUMED (unit amplitudes of the two states).

τ_{SUI} old; uniform random activation (URAP) and a bandit over a codebook of exact optima, with one power report per τ_{SUI}, need no position. Hypotheses and ranges preceded the sweep; the committed decision rules followed it (Appendix F). The points come not from the engine's `run_batch` but from an offline vectorized evaluator of the same model (channel on a grid of 1 mm or finer, decisions on a 1 cm grid, each 5 ms step evaluated at its midpoint, a rule adopted after a short IQ run), over 20 batches of 500 mean movement periods of one trajectory.

With instantaneous position knowledge the exact optimum gains 5.61 dB over RIS absent at 1 GHz (95 % interval 5.53–5.69 dB, mean of the per-batch dB gains), and 5.48 dB at 2.4 GHz. We measure the decay with delay (Fig. 5) by the retention $\rho(\tau) = (P(\tau) - P_{\mathrm{stale}})/(P(0) - P_{\mathrm{stale}})$ in linear power, where P_{stale} applies at each step the exact optimum for the position half the run away along the trajectory. Half the benefit is lost at $\tau_{1/2} = 0.126$ s (95 % interval 0.125–0.127 s), and at 2.4 GHz at $\tau_{1/2} = 0.054$ s. In units of λ/v the two half-lives are 0.421 and 0.433: the time scale is λ/v, as the classical Doppler decorrelation of a moving terminal predicts [72], [73], not the mobility period, and outdated channel knowledge is known to erode RIS gains under mobility [74], [75]. Both intervals lie inside the wide ranges fixed before the sweep, a weak test of H-A1 and H-A2. The constant is not universal: over the 12 geometries computed (Table XII), $\tau_{1/2}$ spans 0.27–0.55 λ/v, rising with the partition loss and falling with the offset of the cells, for one trajectory at one speed. Without the partition, or with one acting on both paths (Appendix F), the optimum gains 0.68 dB at 1 GHz, with $\tau_{1/2} = 0.106$ s. Without positions, the bandit beats URAP at every delay from 0.3 s by 0.38–0.39 dB (H-A3) but stays near the stale level (the bandit gains 0.50 dB, the stale configuration 0.50 dB).

As a consistency check, not a finding, 8 runs of 1000 s through the served engine, with an external controller in barrier mode and an external receiver application (one geometry, exact optimum and URAP, delays of 0.1 and 1 s, mean power alone), agree with the evaluator within 0.002 dB, inside the pre-specified limit of 0.02 dB (VA-1, Fig. 5(c)). The engine's `run_batch` reads up to 0.09 dB higher and fails that limit (Appendix F).

C. Non-ideality ablation

Because the optimum over a finite alphabet is exact (Section IV-D), the cost of a non-ideality, the loss of the optimized RIS term relative to ideal continuous phase, is a short analytic result rather than the outcome of a heuristic (Appendix F); the results agree with prior work [8], [11]. One-bit control costs 2.9–3.3 dB, less than the large-N closed-form loss of 3.92 dB, because a finite panel in a fixed geometry is not a random phase ensemble. The laboratory states, with magnitudes ASSUMED to be unit, are nearly antipodal (in $\Gamma = m \pm d$, $|d|$ is -0.18 dB and $|m|$ is -13.9 dB) and behave like ideal one-bit states (within 0.10 dB). Unequal amplitudes

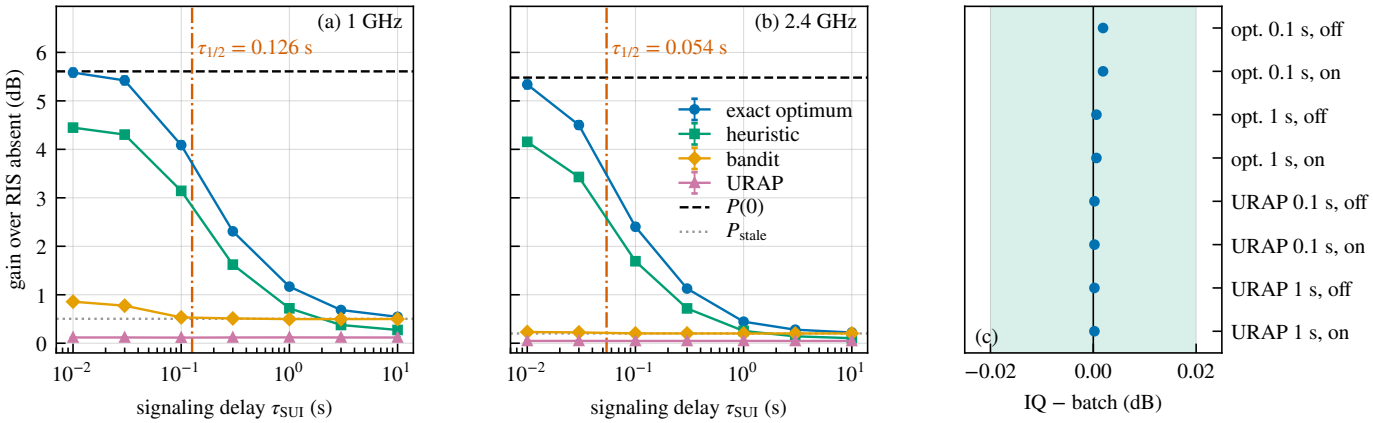


Fig. 5. Signaling delay in closed-loop binary RIS control in the blocked corridor. (a), (b) Gain over RIS absent versus the signaling delay (the age of the position reports; for the bandit, the interval of its power reports) at the two carrier frequencies for four controllers (mean of per-batch dB gains with intervals), with the instantaneous optimum $P(0)$, the stale reference and the half-retention delay $\tau_{1/2}$ of the exact optimum. (c) IQ path minus the E-A evaluator for the consistency runs (rows: controller, delay, noise), with the acceptance band. Data class: SIM-sremu.

cost up to 5.2 dB when one state reflects at half amplitude, an efficiency of one half adds about 3.0 dB, and designing for an ideal alphabet but deploying on the true states costs 0.16–0.44 dB more than designing for them.

D. Integration with external IQ applications

The raw request–reply mode (Section III-C) ran with unmodified GNU Radio 3.10.9.2 (Ubuntu package) as the peer, in a separate process that knows nothing of the emulator: its stock ZMQ request and reply blocks and a differential QPSK (DQPSK) modem of stock blocks with blind timing and carrier recovery. In the scene (SIM), transmitter and receiver stand 5 m from an ideal 1-bit RIS of 256 cells at 3.5 GHz, with a partition of 40 dB on the direct path, at 1 MS/s (Appendix F). In 10 fresh runs of 0.3 s, one per RIS configuration, GNU Radio makes no symbol error with focusing, its error rates do not decrease as the configuration's gain falls, and the random configuration loses every block; the gains lie within 1.0 dB of an array-factor prediction, an internal consistency check, not a measured RIS gain.

A further run switched the RIS three times, 0.4 s apart, by commands scheduled over the control socket before virtual time started: each switch was applied with zero lateness, and the change point of the IQ that GNU Radio received lies at the logged switch sample (largest offset 0 samples). GNU Radio received every sample, bit for bit the stream that the emulator recorded, and across two independent repetitions of all 11 runs the SHA-256 digests of the received IQ and of GNU Radio's decisions are identical. The runs are lockstep, so the sample exchange paces the emulator and the wall-clock speed changes how long a run takes, not its samples. srsRAN and OpenAirInterface were analysed but not run (Appendix F).

VIII. Discussion

A. Alignment with standards

`sremu` keeps the split of ETSI GR RIS 002 [5] between the RIS controller, which generates control information, and the micro-controller, which configures the elements (here the engine, at the sample of each receiver). The general-case path-loss model of GR RIS 003 [6] equals K1 for one point per cell, and the reconfiguration time of GS RIS 008 [7], from the receipt of a command to the effective configuration, is in `sremu` the logged command latency plus the transition. Appendix A maps the ETSI terms to `sremu` as alignment, not conformance; the E2 service model for RIS of O-RAN frameworks [26], [27] is not implemented.

B. Hooks for learning-based control

Learning-based control is outside our scope, but `sremu` provides three hooks: one controller object runs in the engine or in an external process with identical samples, with built-in baselines (Appendix B); an environment, `RisEnv`, wraps a scenario in the reset/step interface of Gymnasium [76], with rewards computed on the batch path in this release; and at every step it reports the exact finite-alphabet optimum (Section IV-D) from oracle channel state and the gap to the achieved power, a regret reference that bounds every built-in controller at every tested position (VA-3).

C. Limitations

Appendix C lists what the release does not model or provide. Four limits matter most here. The paced mode is soft real time: it counts deadline misses and does not prevent them. No timing, switching or transient behaviour has been compared with measurements, so E-B is model-based, and the data of Tang et al. test the absolute level of one static state (Section V). Mutual coupling is not modelled beyond what a per-state table with identical neighbours holds, so a controller cannot exploit coupling between differing states. And occluders never attenuate a RIS hop, so the blocked arc of E-B and the corridor of E-A are idealized.

IX. Future Work

Validation. The next validation step is the open sub-6 GHz data set of Rossanese et al. [62]: beam scans of a surface

with a documented chamber geometry. Its comparison needs a new analysis plan, locked before the emulator is run on the data: the measured columns sit on a floor that does not depend on the configuration, the cells have seven phase states and an absorbing state [63] rather than an ideal alphabet, and the descriptive structure of the data has already been seen, so the comparison can be pre-specified but not blind. A measured reconfiguration transient would move parts of the model card from consistency checks to validation. *Unmodified radio software.* A virtual-USRP layer of the kind ACHEM provides [33] would let unmodified UHD applications use the emulator, and srsRAN would need a lockstep peer that offsets the transmit and receive streams its ZMQ driver [32] ties together (from reading the driver, not tested; Appendix F). *Network control.* An E2 service model adapter for RIS [26], [27] would connect O-RAN applications, and authenticated sockets would allow control across hosts. *Physics.* Candidates are mutual coupling through a mutual-impedance model [51], RIS–RIS paths, frequency-dependent reflection filters for signals wider than a table's validity bandwidth, polarization and non-reciprocal tables. *Runtime.* Candidates are rate conversion between streams, a pipeline resident on a graphics processing unit (GPU) for large arrays, a transport outside the engine's interpreter for the served data plane, and vectorized learning environments with IQ backends.

X. Conclusion

In `sremu` a RIS command takes effect at a requested and logged sample on every receiver, and lockstep runs with barrier controllers, in process or over the control socket, are deterministic. With the line of sight blocked, the packet error rate followed the closed form of the semantics: commands applied on arrival lost no packet while their worst latency stayed below the guard plus the 0.256 ms preamble, timed ones below their lead plus half the guard plus the preamble; with a line of sight, the waveform set the cost. Position reports must be younger than a fraction of λ/v, as Doppler decorrelation predicts [72], [73]: half of the RIS benefit was gone after 0.27–0.55 λ/v across geometries (0.421 λ/v in the primary one). Designing for the laboratory states rather than an ideal alphabet recovered up to 0.44 dB in the geometries studied (magnitudes assumed).

Served on one AWS host at 10–100 commands/s, at least 99.8 % of commands timed 4 ms ahead took effect at their exact sample, though page deadlines failed our Sustainable rule; leads must cover the measured latency of the controller's path. A laboratory comparison under a pre-specified plan hash-locked before unblinding was underpowered and did not separate the RIS model from LOS-only; at 3.5 GHz a RIS map needs its geometry recorded to about a millimetre. Third-party chamber data test the model class, whose published offset the emulator inherits; no switching or timing behaviour was compared with measurements.

Code and Data Availability

The emulator `sremu` is available at https://github.com/Galabavamsi/sremu under the MIT licence. The pipeline that builds this paper from recorded results (sweep specifications, result files with their provenance, figure scripts and checks) and the paper sources are at https://github.com/Galabavamsi/sremu-paper under the same licence; `sremu.lock` names the emulator commit the results come from. The analysis plans of Sections V-D and VII-B, their lock records and the digitized laboratory map of Section V-D are in the same repositories. Third-party values, such as the curves and points of [36], were digitized from published figures and are kept in the emulator repository with their source and digitization record.

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This work builds on the author's earlier emulator for smart radio environments, written during a 2025 research internship at IIT Bhilai on a COMET Foundation-sponsored project on smart radio environments advised by Dr. Arzad Alam Kherani: a time-stepped Python link-level emulator with node mobility, line-of-sight and RIS channel functions, a USRP-style IQ interface and external traffic interfaces (https://github.com/Galabavamsi/RIS-SIM), its 2026 rewrite with a browser dashboard and scenario library (https://github.com/Galabavamsi/RIM-SIM-V2), and the conference manuscript built on them, an unpublished ANTS 2025 submission with S. Mishra, A. Jaiswal, S. Kumar, M. Ambwani and A. A. Kherani. The problem statement, the requirements (sample-level timing of RIS commands, an SDR-style interface, controllers in the loop), the laboratory and FM scenarios and the reviewer feedback that shaped this paper come from that work. The author thanks Dr. Kherani for advising it and the co-authors of the earlier submission, whose laboratory measurement is used in Section V-D. The runs marked AWS in Appendix I used Amazon Web Services EC2 instances.

Use of generative AI. `sremu`, its tests and analysis scripts, the paper pipeline and drafts of this manuscript were written with substantial assistance from Anthropic's Claude (Claude Code), working under the author's direction; the author set the scope, the requirements and the decisions recorded in the paper. Every reported number is produced by code from recorded results and is checked by automated tests, independent analytic and MATLAB references and recorded provenance (Section V, Appendices D–I); a review panel emulated with the same assistant preceded the final revision. The author takes full responsibility for the content.

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APPENDIX A
TOOL SURVEY AND STANDARDS ALIGNMENT

Table II gives the full comparison behind Table I: versions and dates, the level of each tool, wall-clock behaviour, RIS model, measured responses, mobility, external RIS control and licence. The survey of Section II-F covers tools published or released up to 10 Oct 2026, found through Crossref and arXiv searches, the reference lists of the papers found and the repositories and documentation they link; a tool entered the comparison when it models a RIS or exchanges IQ samples with radio software, each feature was read from a primary source (paper, documentation or code), and a tool counts as open when its code is publicly downloadable.

Some details of the tools of Section II complete the table. Wireless InSite describes engineered electromagnetic surfaces with a uniform-ray physical-optics model for a static configuration [23], [24], and a ray-traced model of a deployed surface has been used to choose its phases and checked against measurements [49]. The ns-3 modules model 5G and unmanned-aerial-vehicle links [54], [55], and ns-3 can draw its channels from Sionna RT, whose current releases have no RIS [56]. The hardware demonstration of [57] closes the loop through the RAN intelligent controller. The Colosseum channel emulator is built on field-programmable gate arrays (FPGAs) [30]. Software-in-the-loop emulators couple the OAI rfsimulator with ray-traced channel taps [77] or run GNU Radio channel models for swarms of unmanned aerial vehicles [78], and NVIDIA's ray-traced RAN simulation platform (AODT) has no RIS files in its repository [79].

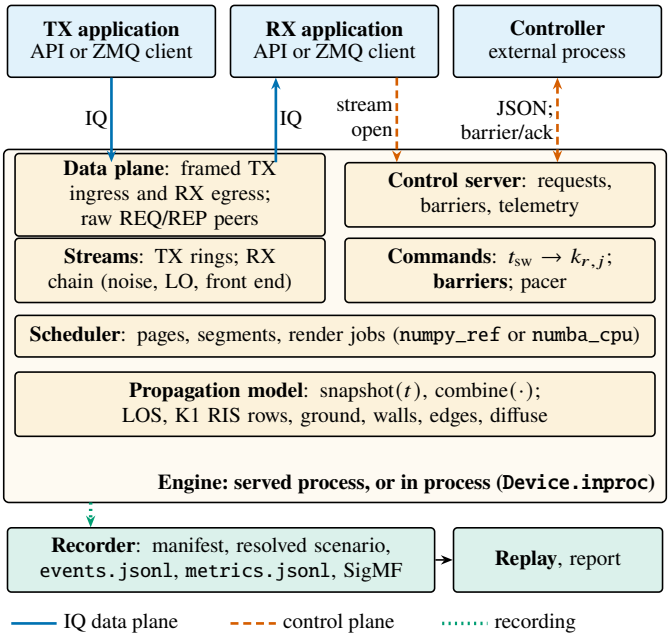


Fig. 6. Architecture of `sremu`. Applications and controllers are separate processes (or run in the engine's process through `Device.inproc`). Solid lines: IQ data plane; dashed: control plane; dotted: recording.

Table III maps the terms and models of the ETSI documents on RIS to elements of `sremu`. In the split of GR RIS 002 [5], controllers decide, and the engine applies their commands with the panel's control granularity, quantizer and failure map at the sample of each receiver. The reradiation integral of GR RIS 003 [6] has the factor j that K1 keeps; its mutually coupled antenna model and its polarized elements are outside the model of `sremu`. The parameters of GS RIS 008 [7] appear as fields of the response tables and of the panel capabilities (Appendix B), and the default duration of the transition is the panel's reconfiguration time. The modulated-signal quality of GS RIS 008, the error vector magnitude (EVM), is measured by an RX application on received samples; Section VII uses it.

APPENDIX B
IMPLEMENTATION AND INTERFACE DETAILS

This appendix extends Section III and the hooks of Section VIII-B. Fig. 6 shows the processes of `sremu` and the planes that connect them. The package is built on NumPy and SciPy, with pyzmq for the sockets [64] and pydantic for validation; Gymnasium and SigMF are optional [65], [76].

TABLE II
RIS SIMULATORS, RAY TRACERS, CONTROLLER FRAMEWORKS AND IQ-LEVEL EMULATORS, AS OF 10 OCT 2026. "NV": NOT VERIFIED FROM A PRIMARY SOURCE EITHER WAY; "NO EVID.": A PRIMARY SOURCE WAS READ AND DOES NOT MENTION THE FEATURE; "NO": THE SOURCE STATES THE FEATURE IS ABSENT. COEF.: CHANNEL COEFFICIENTS; CIR: CHANNEL IMPULSE RESPONSE; TS: TIME-SERIES COEFFICIENTS; IQ: COMPLEX-BASEBAND SAMPLES; DES: DISCRETE-EVENT SIMULATION; PO: PHYSICAL OPTICS.

Tool (version, date)	Level	Wall-clock	RIS model	Measured responses	Mobility	External RIS control	Open; licence
SimRIS v2.0 (2020; code 2023) [15], [16]	Coef. (Monte Carlo H, G, D)	No	Statistical mmWave cascade, ideal phase	no evid.	no evid.	No	Yes; MATLAB; GPL-2.0
QRIS (2023) [17], [18]	TS (QuaDRiGa)	No	Unit-cell patterns on QuaDRiGa geometry, interacting surfaces	Yes (unit-cell patterns from measurement or CST)	Inherited from QuaDRiGa; QRIS-specific nv	no evid.	On request; licence not stated
Sionna RT: v0.18–0.19 (RIS module); v1.0–2.2.0 (29 Sep 2026) [19], [20], [52]	Coef./CIR, radio maps	No	v0.18–0.19: macroscopic reradiation, per-cell profiles. v1.0–2.2.0: none	v0.18–0.19: partial (spatial profiles; state table nv)	nv	v0.18–0.19: gradient-based optimization	Yes; Python; Apache-2.0
MATLAB R2025a+ (checked in R2026a) [21], [22]	Coef. and samples, offline	No	Element-pattern unit cell; CDL cascade per GR RIS 003	Partial (tabulated element pattern; bistatic state table nv)	Via user script	User script	No (commercial)
Wireless InSite EES (Dec 2022) [23], [24]	Ray paths, coverage	No	Phase-gradient metasurface (uniform-ray PO), single static configuration	nv	nv	No	No (commercial)
CoopeRIS (2024) [25]	Packets (DES; SINR/MCS)	No (DES)	Far-field per-element phase sum	nv	Yes (SUMO vehicles)	Internal oracle controller	Yes; C++; GPL-2.0
GoSimRIS (2025; code public 31 Mar 2026) [26]	Coef. + SNR/rate	E2 exchange (author claim)	Configurable phases, at most 64 elements	no evid.	Position list input; model nv	Yes (FlexRIC xApp, E2SM-RIS)	Yes; Go; Apache-2.0
ORIX / E-GoSimRIS (2026, early access) [27]	Coef.	Near-RT loop, 10 ms–1 s	Quantized phase, phase-dependent amplitude	Parametric fit (not a table)	3GPP InF temporal dynamics	Yes (E2SM-RIS)	nv
Colosseum + RIS scenarios (2025) [28]–[30]	IQ (FPGA FIR, 256 SDRs)	Yes (hardware)	Precomputed QuaDRiGa cascade, phases fixed offline	No	Precomputed taps	RAN xApps: yes; RIS during a run: no	No (facility)
OAI rfsimulator (develop, accessed 10 Oct 2026) [31]	IQ (TCP)	No (by design)	None; run-time beam-gain table	No	Channel model via telnet; trajectories nv	Telnet channel control	Yes; C; OAI CSSL v1.0
srsRAN 4G ZMQ (accessed 10 Oct 2026) [32]	IQ (ZeroMQ)	Software timing; nv	None (user flowgraph)	No	User-built	User-built	Yes; C++; AGPL-3.0
ACHEM (arXiv v1, Apr 2026) [33]	IQ + SDR API (unmodified UHD applications)	Yes (benchmarked)	None	Antenna patterns; no RIS	Yes (ArduPilot SITL)	Online channel reconfiguration; no RIS	Yes; licence nv
TRACE (arXiv v1, Aug 2026) [34]	Symbols	nv; UDP control RTT 5.2 ms reported	Ideal continuous phase, unit amplitude	No	Channel evolution (random walk, Gauss–Markov)	Yes (UDP control plane)	Code "upon acceptance"
`sremu` (this work)	IQ + SDR-style API (UHD-shaped)	Lockstep: sample-exact; paced: measured (Section VI)	Area-tied aperture kernel; per-state tables $\Gamma(s, f, \theta_i, \theta_s)$	Import path tested (state, frequency, both angles in one plane of incidence, provenance); no measured per-state table used here (reported and assumed states)	Analytic trajectories, geometric Doppler	Yes (timestamped, applied and logged per receiver)	Yes, MIT licence (see Code and Data Availability)

A. Recording, time base and command bookkeeping

A recorder writes one directory per run: a manifest (software versions, git commit and dirty flag, platform, seed and random-number scheme, hashes and data classes of the inputs, mode, backend, per-link interpolation and filter choices, loss counters), the resolved scenario, an event log with one JSON line per change of a command record, flushed as it is written, a metrics log per 100 ms of virtual time and, optionally, SigMF recordings [65] of the streams whose annotations mark each applied command at its sample. A replay tool and a static HTML report read run directories. The control plane carries stream management, RIS commands, controller registration and session control. Registered controllers receive their barrier messages on the control connection, never on the publish socket, which may drop messages. A served session can start paused, the default of the serving command, so that clients can open streams and issue timed commands before virtual time advances.

With $\mathrm{to_ticks}$ an event lands within $1/(2f)$ of its requested time. Floating-point times from the Python interface are converted through their shortest decimal representation. Each RX stream r is processed in pages of $P_r = \mathrm{clamp}\big(2^{\mathrm{round}(\log_2 f_r T_\mathrm{p})}, P_\mathrm{min}, P_\mathrm{max}\big)$ samples on an absolute grid, with a page target T_p of 0.5 ms by default, $P_\mathrm{min} = 64$ and $P_\mathrm{max} = 16384$; streams of equal rate and page size share one cursor. Per-sample quantities depend on the absolute sample index and the two enclosing snapshots, recursions restart from closed-form anchors at page starts and snapshot boundaries, parallel work splits samples, and the RX samples do not depend on barrier limits, input availability or whether inactive

TABLE III
ALIGNMENT OF SREMU WITH THE ETSI DOCUMENTS ON RIS. A DASH IN THE LAST COLUMN MARKS A CLAUSE THAT SREMU DOES NOT MODEL. THE MAPPING STATES ALIGNMENT OF TERMS AND MODELS; IT IS NOT A CONFORMANCE CLAIM.

Document	Clause and term or parameter	Element of the emulator	Evidence
GR RIS 002 V1.2.1	§3.1, §4.2: RIS controller, which generates the control information; five controlling types [5]	External or in-process controller (one protocol; barriers in lockstep); the controlling types are not modelled separately	RT-08, IO-03
	§3.1, §5.1: micro-controller, which configures the elements [5]	Panel command logic: control granularity, quantizer, failure map, switch per receiver	RT-05, PH-30
	§5.1: control signalling from controller to RIS [5]	Control socket; latency model (fixed plus keyed jitter); command records with lateness	RT-06, RT-17
GR RIS 003 V1.2.1	§5.1.2: locally periodic discrete model; alphabet of reflection coefficients [6]	Response tables $\Gamma(s, f, \theta_i, \theta_s)$ with provenance and validators	PH-31
	§5.1.3: mutually coupled antenna model [6]	Not modelled; per-state tables hold coupling for identical neighbours (§5.1.2); a hook protocol, implemented as no coupling	—
	§6.3.1.2: path-loss model, general case, with the recommended cell gain [6]	K1 with one point per cell is algebraically equal to it	PH-02
	§6.6: reradiation integral with factor j and point-wise Γ [6]	Factor j of K1; `po_patch` quadrature of the cell integral; one reradiated mode per state	PH-05, PH-26
	§6.8: polarized element modelling [6]	Not modelled (scalar co-polar fields)	—
GS RIS 008 V1.1.1	§4.1, §4.2: operating frequency range; maximum operating bandwidth [7]	Frequency axis and validity bandwidth of a table; capabilities `operating_band_hz`, `max_bandwidth_hz`; wider signals rejected	PH-31
	§4.3, §4.4: bistatic radar cross section; reflected-beam spatial characteristics [7]	Incidence and reflection angle axes of a table (one plane of incidence); beam patterns compared with MATLAB (code to code)	G2, G3
	§4.5: total reflected power [7]	Energy diagnostic: scattered power at most the intercepted power	PH-23
	§4.6: modulated-signal quality (EVM) [7]	EVM measured by an RX application on received IQ samples (Section VII)	F9
	§4.7: static configuration power consumption [7]	Capability `static_power_w`, kept as metadata	—
	§4.8: reconfiguration time, from command receipt to effective configuration [7]	Command latency (latency model, measured lateness) plus the transition, whose default duration is `reconfiguration_time_s`	RT-12, RT-17

pages were skipped (RT-03, RT-10). A change of the page target is not bit-exact: it changes the double-precision accumulators within the relative tolerance of RT-04 (Section V). Deterministic mode, the default of the command-line tools and the mode recorded by every emulator run reported here, compiles the kernels without Intel's short vector math library and keeps their cache apart, so that how parallel work splits the samples cannot change a result; without it, thread-count invariance is not claimed bit for bit.

A command record also holds the command's identifiers, and the configuration hash is a BLAKE2b hash of the applied configuration. Every command is kept: of two configuration commands for one panel at one time, the later sequence number wins and the earlier is recorded as superseded (RT-07). Switching a panel off and on ('RIS absent') is a separate timeline with the same per-receiver switch samples; its records are flagged as counterfactual, and it never changes element states. The transition profiles are a step, a linear ramp over T_{tr} or $1 - e^{-t/T_{\mathrm{tr}}}$; the complex-linear path between states stays an engineering choice until a measured transition is available. A command that arrives during a transition starts from the instantaneous reflection coefficients.

The observation of a barrier controller holds the power measured from each receiver's IQ samples over a window and the RIS state, plus, on request, node positions (optionally delayed and perturbed), an IQ window, or oracle channel state, whose use the manifest flags. A remote controller receives `ctrl.barrier`, sends its commands and replies `ctrl.ack`. The batch path applies switches at their surface time because surface-to-receiver delays are negligible at its level, and it reports those delays.

In lockstep mode, after 2 s of wall time without progress the engine writes a stall report (RT-13). In paced mode the input slack D_{lag} is 1 ms by default, and the pacer sleeps and then spins for the last 1.5 ms before a release. Late timed TX bursts are dropped with a time error, missing untimed samples become zeros with an underflow report, an RX client that falls behind loses pages with an overflow report, and deadline misses are counted at slacks of 0, 1 and 5 ms (RT-18). Lockstep mode with asynchronous controllers is rejected.

B. Python API and wrappers

Table IV lists the Python API of `sremu` next to its UHD counterparts [37]. TX and RX streams carry UHD-style metadata with exact time specifications, RX streams follow the UHD stream modes, and timing and protocol failures are status codes: the UHD receive and asynchronous codes plus codes for schema errors, lookahead violations, unknown targets, out-of-domain queries, oversize messages, refused operations and rate mismatches (IO-06). Phases in a RIS command are

TABLE IV
PYTHON INTERFACE OF SREMU (UHD-SHAPED, NOT UHD-COMPATIBLE), ITS UHD COUNTERPART [37] AND THE TWO ONE-CALL CONVENIENCE WRAPPERS FOR SIMPLE SCRIPTS. GENERATED FROM THE SREMU.API.METADATA DOCSTRING.

sremu API	UHD	Wrappers Send_to_sim / Receive_from_sim
TxStreamer.send(buf, TxMetadata, timeout)	tx_streamer::send	Send_to_sim(node, samples, fc, fs, gain, start_time): one burst (start and end of burst)
RxStreamer.recv(buf, RxMetadata, timeout)	rx_streamer::recv	Receive_from_sim(node, nsamps, fc, fs, gain, start_time): exactly nsamps samples
TxMetadata (SOB, EOB, time_spec; sremu: idle_until, end_of_stream)	tx_metadata_t	set by Send_to_sim; start_time makes the burst timed
RxMetadata (time_spec, error_code, EOB; sremu: first_sample)	rx_metadata_t	none; error codes become exceptions
TxStreamer.recv_async_msg(timeout) returning AsyncMetadata	tx_streamer::recv_async_msg, async_metadata_t	none
RxStreamer.issue_stream_cmd(StreamCmd)	rx_streamer::issue_stream_cmd, stream_cmd_t	issued by Receive_from_sim (num_done)
TimeSpec(full_secs, frac_secs) (exact; floats via Fraction(repr(x)))	time_spec_t	start_time in seconds (float, exact string or TimeSpec)
Device.set_command_time / Device. clear_command_time	multi_usrp::set_command_time / multi_usrp::clear_command_time	none
StreamArgs(stream, fc_hz, rate_sps, cpu)	stream_args_t with set_tx_freq/set_rx_freq, set_tx_rate/set_rx_rate	fc, fs: open <node>.tx0 or <node>.rx0 at the opening call
TxStreamer.idle_until(TimeSpec) (sremu extension)	none (closest: end of burst)	none
Device.ris(panel).set_config(...) (timed RIS command)	none	none

quantized by the panel, and set_command_time times later commands as in UHD. Two wrappers, Send_to_sim and Receive_from_sim, give simple scripts a one-call interface; a BPSK transmitter and receiver script pair delivers every sample through them (IO-07).

C. Data plane, control plane and security

A data-plane message has a 48-byte little-endian header and a payload sent without copying. The header holds a magic word, version, flags (start and end of burst, timed, idle-until, gap before, late, end of stream), stream identifier, a per-stream sequence number whose gaps raise a sequence error, the absolute index of the earliest sample, the sample count (at most 2^{22}), the channel count, the sample format (cf32_le or sc16_le) and the sender's monotonic clock, which feeds latency metrics and never the data plane. Malformed frames are dropped and counted, and no input of the fuzzed corpus made the engine raise (IO-01); TX sockets block at their high-water mark, which gives lockstep backpressure.

Control messages are JSON objects (envelope version 1) validated before dispatch in a fixed order: size (at most 1 MiB), syntax, element count (at most 65 536 per command) and schema, so every refusal has one code and is counted (IO-01, IO-04); a message above the transport limit of 16 MiB closes the connection without a reply. The operations cover version negotiation, panel capabilities, frontier time, RIS configurations, codebooks, the counterfactual enable state, node positions and trajectories, streams, controller registration, acknowledgement and removal, recording markers, statistics, and pause, resume and shutdown; a socket command lands at its exact sample with zero lateness in lockstep mode (IO-03). No message field is interpreted as a file path, sockets bind to loopback addresses unless remote binding is allowed, there is no authentication, and commands above a per-client rate limit (10 kHz by default) are refused and counted (IO-04).

D. Response tables and panel capabilities

One angle grid serves both angles of a table, and tables without angle or frequency dependence are special cases. The provenance records the kind of source (measured, full-wave, circuit, analytic, digitized or assumed), the reference, reference plane, polarization, whether the cell pattern is included, magnitude and phase standard deviations, a hash of the raw data, licence, creation record and every transformation applied after loading. Validators reject a table that is not passive ($|\Gamma|$ above unity beyond a tolerance that grows with the stated uncertainty of measured and digitized data), not reciprocal ($\Gamma(s;\theta_i,\theta_s) \neq \Gamma(s;\theta_s,\theta_i)$ beyond a tolerance derived from the stated deviations; recorded symmetrization is available) or too coarse (adjacent entries differ by more than 0.35 of the largest magnitude) when it is loaded, and they reject a query outside its grid and a linked signal wider than its validity bandwidth (PH-31). Interpolation is complex-linear in angle and linear in frequency, never in magnitude and unwrapped phase. A table that depends on incidence alone enters through the recorded reciprocal extension $\Gamma(\theta_i,\theta_s) = g(|\theta_i - \theta_s|/2)$ (PH-28); raw non-reciprocal tables are rejected.

Tables are imported from long-form CSV files with a provenance sidecar, or from MATLAB unit-cell phase curves labelled as simulated stand-ins. The built-in tables are the two-

state laboratory alphabet, whose unknown amplitudes must be set and are recorded as assumptions; the published states of the two millimeter-wave surfaces of Tang et al. [36]; ideal b-bit and antipodal alphabets; the on–off alphabet of [13]; and the phase-dependent amplitude model of [11]. A table can draw perturbed copies from its stated uncertainty for Monte Carlo studies.

Panel capabilities are named after the parameters of GS RIS 008 [7] where such a parameter exists: number of elements, layout, control granularity (element, column, tile or groups), state labels, source of Γ (measured with the data hash, or analytic with the model name), operating band, maximum bandwidth, reconfiguration time, maximum command rate and static power, kept as metadata. Table III (Appendix A) maps these fields and the kernel to the ETSI documents as alignment, not conformance.

E. Controllers and learning environment

A controller has a name, a period, a minimum latency, an observation specification and a latency model, and two methods: `setup` receives the panel capabilities and a keyed random generator, and `decide` maps an observation to actions. The same object runs inside the engine at barriers or in an external process (`run_remote`), and both give identical received samples (RT-08), so a controller tested in process can be deployed as a separate process without changes. The built-in controllers are a fixed configuration; RIS absent; the uniform random activation and our reading of the position-driven heuristic of [13]; the exact optimum; phase alignment followed by quantization; a design for an ideal alphabet deployed on the true states; a cyclic codebook sweep; an upper-confidence-bound (UCB1) bandit [80] over windowed received power; a genetic algorithm; a controller that alternates configurations between the packets of two receivers; and a wrapper that delays the position reports of any position-driven controller. Every random draw of a controller comes from its keyed generator, so its decisions are reproducible.

`RisEnv` subclasses Gymnasium's environment class when that package is installed and imitates it otherwise. An action is a state vector or a codebook index, applied after a configurable latency; the reward is the received pilot-tone power over the decision period; the samplers of the action and observation spaces and the scene are keyed by the reset seed. Its IQ backends are not yet connected to the engine. The optimum is computed at the same evaluation times as the achieved power. The regret gap is zero for an optimal controller; it ignores panel failures, so it can be negative when cells are dead.

APPENDIX C
PHYSICAL-MODEL DETAILS

This appendix extends Section IV. Table V lists every component of the model with its validity domain, verification evidence and default; the identifiers refer to Section V and Appendix D.

A. Conventions and line of sight

The trajectories are static, piecewise-linear waypoints, constant velocity, one- and two-dimensional random waypoint [81] with keyed legs, a cubic-spline trace, or timed updates from a client, so positions never depend on an update grid. Tables without angle dependence use a separable form of the coefficient tensor. Tables with angle dependence are cuts in one plane of incidence: a direction $\mathbf{d}$ from a cell enters as the signed angle $\operatorname{atan2}(\mathbf{d}\cdot\mathbf{u}, \mathbf{d}\cdot\mathbf{n})$ in the plane of the panel normal $\mathbf{n}$ and its axis $\mathbf{u}$; θ_i uses the direction from the cell to the transmitter and θ_s the direction from the cell to the receiver, both outward, so specular reflection is $\theta_s = -\theta_i$ in the table schema $\Gamma(s, f, \theta_i, \theta_s)$, and its component along the other axis is projected out (a panel option refuses off-plane geometry instead), while the obliquity and the front-side indicator of K1 use the full cosines. With this convention the reciprocal extension of a table of the incidence angle alone (Appendix B, PH-28) returns $g(|\theta|)$ at specular reflection and $g(0)$ at retro-reflection, an engineering choice (ENG). A table is read at the carrier of the link (linear in frequency between its grid points) and held over the signal's band, which may not exceed the table's validity bandwidth; there is no Γ filter. No measured angle-dependent table is used in this paper. Channel coefficients are dimensionless and normalized as in the Friis equation, $P_R = P_T|h|^2$, with antenna gains inside h. The complex baseband signal $x(t)$ of a carrier f_c is $s(t) = \sqrt{2}\,\mathrm{Re}\{x(t)e^{j2\pi f_c t}\}$, and a path with delay τ maps $x(t)$ to $a\,x(t-\tau)\,e^{-j2\pi f_c \tau}$.

The LOS path is $h_{\mathrm{LOS}} = \sqrt{G_T G_R}\,\lambda/(4\pi d)\,V$ with scalar co-polar antenna patterns (isotropic, a $\cos^{\alpha}$ pattern with peak gain $2(\alpha+1)$, or a tabulated pattern with provenance) and an explicit visibility factor V: zero behind an opaque occluder, $10^{-L/20}$ through a partition with transmission loss L dB (an engineering choice), or the Fresnel knife-edge factor $F(\nu)$ of one edge, which keeps its phase and adds coherently to the RIS paths. Occluders and partitions act on the LOS path and on the ground and wall images, never on a RIS hop, so a blocked scenario whose occluder also crosses RIS hops is idealized (Appendix F). Scenario names never imply blockage. Optional mechanisms add a flat-ground image path with Fresnel coefficients, image expansion of the two RIS hops over that ground, order-one images in finite walls and an additive diffuse term (Table V).

B. Kernel guards, cell models and non-idealities

A scenario check warns when a cell is closer to a node than three wavelengths, where the point-scatterer approximation weakens, and rejects distances below one wavelength. With the default obliquity the scattered power of a passive panel does not exceed the power it intercepts in any tested configuration (PH-23); the alternative obliquity $(\cos\theta_i + \cos\theta_s)/2$ can exceed it for anomalous reflection and therefore emits a validity warning when selected. With that Kirchhoff obliquity K1 is the physical-optics integral of the aperture; with the default it is an area-tied aperture kernel whose obliquity follows the measured cosine pattern, and both agree at specular reflection. The

TABLE V
MODEL CARD OF SREMU: MODEL, VALIDITY DOMAIN, VERIFICATION EVIDENCE AND DEFAULT OF EACH COMPONENT. GENERATED FROM DOCS/MODEL_CARD.YAML; EVIDENCE IDS REFER TO THE TEST TABLES OF SECTION V. ENG MARKS AN ENGINEERING CHOICE WITHOUT A LITERATURE SOURCE.

Component	Model	Validity domain	Evidence	Default
LOS	Friis + patterns + explicit V	Far field of the antennas; occluders act on LOS (and ground/wall image legs), never on RIS hops	`PH-01`	On
RIS	K1, `point`	Point-scatterer regime of Tx and Rx ($r \geq 3\lambda$); pitch $\leq \lambda/2$; scalar co-pol	`PH-02-07`, `G1-3`, `V-TANG`	auto
RIS	K1, `po_patch`	As above; uniform-patch PO; Richardson-converged	`PH-04`, `PH-26`	Pitch $> \lambda/2$
Γ tables	Complex-linear LUT	Inside the grid; band $\leq$ validity bandwidth; reciprocal	`PH-28`, `PH-31`	—
Ground	Flat-earth image + Fresnel	Smooth, flat; no curvature (P.526 experimental)	`PH-10`, `PH-32`, `G8`	Off
Walls	Image, order 1	Finite polygons; specular	`PH-08`	Off
Diffraction	Single knife edge $F(\nu)$	One dominant edge	`PH-11`	Off
Diffuse	Additive SoS	Statistical background only	`ST-02`, `G6`	Off
Mobility/Doppler	Chord delay + exact phase + tiles	$\varepsilon_{\mathrm{rel}} \leq 10^{-3}$ (bound plus `verify_interp`)	`PH-17-19`	Auto
Wideband	FD Kaiser 8/16/32	$B \leq 0.8 f_s$; ≤ -60 dB in band	`PH-20`	Auto
Noise	$k_B T_0 F f_s$ per RX	—	`PH-21`, `G5`	On
Non-idealities	Ordered pipeline: command, quantizer, failures, table, efficiency, coupling hook, transition	Per-state tables, failures, efficiency; coupling not modelled beyond what a per-state table measured with identical neighbours contains (coupling hook implemented as no coupling)	`PH-13-15`, `PH-25`, `PH-30`	—
Transitions	Per-class g_c, complex-linear Γ path	[ENG] until measured	`RT-12`	Step

continuous physical-optics reference of Fig. 2(a) integrates the same kernel with the default obliquity.

Forcing `point` on cells larger than the `auto` threshold emits a warning, and tables whose measurement already contains the cell pattern force `point` without obliquity, so that the pattern is not counted twice. The two models are not numerically equivalent: a patch carries the sub-aperture factor of a uniform patch, which a point omits. For `po_patch` panels a Richardson check computes the received power of up to 8 representative links at sub-cell size Δ and at $\Delta/2$, halves Δ at most 2 times while the change exceeds 0.05 dB, and fails otherwise; the accepted size and the estimate go to the manifest.

The laboratory pitch of 42.8 mm lies 0.03 mm (0.06 %) below half a wavelength at 3.5 GHz, so `auto` resolves to `point` in E-B, in the laboratory-like geometry of E-C and in E-E, whose cells have the same pitch; the synthetic panel of E-C has a pitch of exactly half a wavelength, which `auto` also resolves to `point`. E-A sets `point` on its quarter-wavelength cells, the geometries of Tang et al. and E-D resolve to it, and the laboratory check draws `point` and `po_patch` with equal prior weight. Forcing `po_patch` on every panel of E-B and of the E-C laboratory geometry (coefficient level; record in Appendix I) lowers the RIS power by 0.2–1.3 dB. The exact optima of E-B change in at most 8 % of the cells, and the `point` optimum loses at most 0.01 dB under `po_patch`; the change of RX1's channel at the E-B1 switch moves by at most 1.5 dB, the most with the line of sight present; and the E-C losses relative to continuous phase move by at most 0.1 dB. In the laboratory check the paired M2 predictions of the two models differ by a median of at most 0.13 dB per ring (95th percentile at most 0.67 dB). For a single half-wavelength cell at 45° or wider retro-reflection the default Richardson check does not converge from sub-cells of 0.2λ and fails, so the value quoted in Section IV-C comes from a finer start.

A command (state indices, phases or a codebook entry) is expanded from controls to physical cells by the panel's control granularity. Stuck and dead cells follow a failure map, and dead cells absorb by default. The efficiency factor scales the amplitude by $\sqrt{\eta}$; it is refused for measured or digitized tables unless explicitly overridden, because their data already contain the loss. The transition of Section III-B acts in time.

For a direction θ the best per-element choice is $s_n(\theta) = \arg\max_s \mathrm{Re}\{Z[s,n]e^{-j\theta}\}$; it can change at the critical angles $\arg(Z[a,n] - Z[b,n]) \pm \pi/2$ and nowhere else, and the optimum is the choice at the direction of the optimal sum, which the sweep over the sorted critical angles finds in $O(S^2 N \log(S^2 N))$ operations for S states and N cells.

C. Interpolation bound and fractional delay

For static scenes snapshots are evaluated at events, and configuration switches are discontinuities at exact sample indices. Between two snapshots the delay of a path without a RIS is a chord, linear in time between its delays at the two snapshots, and its amplitude is complex-linear; the residual of a tile is the sum of its cells' coefficients relative to the tile's reference delay, interpolated complex-linearly. Half of the target $\varepsilon_{\mathrm{rel}}$ goes to each source of error. The chord error of a path with range r is $k|\ddot{r}|T_u^2/8$ with $|\ddot{r}| \leq v_\perp^2/r_{\min}$, which gives

$$T_{\mathrm{curv}} = \sqrt{8(\varepsilon_{\mathrm{rel}}/2)\, r_{\min}/(k\, v_\perp^2)}. \tag{3}$$

The residual of a tile with side D_t turns by $\Delta\phi$, equal to $2\pi v_{\mathrm{rel}} D_t T_u/(\lambda R_{\min})$, per interval, and the midpoint dip of linear interpolation, $\Delta\phi^2/8$, must stay below $\varepsilon_{\mathrm{rel}}/2$, which bounds D_t. The policy starts from $T_{u0} = \min(T_{\mathrm{curv}}, T_{u,\max})$ with $T_{u,\max}$ of 10 ms (a scenario can lower T_u but not raise it), evaluates the periods $T_{u0}/2^m$ for m up to 4 with the tile counts each needs, and picks the pair with the lowest estimated compute cost; the choice goes to the manifest. Each

run checks the bound: in every 64th snapshot interval the engine evaluates the exact snapshot at the interval midpoint and logs $\max |h_{\mathrm{interp}} - h_{\mathrm{true}}|/|h|$ per link; the manifest keeps the maximum, and runs above $2\,\varepsilon_{\mathrm{rel}}$ are flagged (PH-18). Section V compares the rendered Doppler shifts of LOS and RIS paths with closed forms (PH-17).

Each TX stream declares its occupied bandwidth B. A link is narrowband when $\pi B \Delta\tau \leq 0.05$ rad over its delay spread $\Delta\tau$; its paths then share one envelope delay and keep their own exact carrier phases. Otherwise the paths are grouped so that each group passes the same test, and each group gets its own FD filter at its chord delay. The FD filters are Kaiser-windowed sinc filters ($\beta = 8$) [67] with 8, 16 and 32 taps for B/f_s up to 0.2, 0.6 and 0.8, respectively, applied through a polyphase table of 1024 fractional phases with linear interpolation between neighbouring phases. Their in-band error stays below -60 dB for every fractional delay (PH-20); a wider band is rejected. An integer delay without a filter is used when the band is narrow enough for the half-sample error, $\pi B/(2f_s)$, to stay below 0.05 rad.

The thermal noise is complex Gaussian. Random keys are derived with a BLAKE2b hash of the seed, purpose and name, and the counter-based Philox4x32-10 generator [82] takes the absolute index (sample or step) as its counter, so the compiled and reference backends produce identical integer streams (RT-14). Further impairments are carrier-frequency offsets of TX and RX oscillators, computed as phases of the absolute sample index (RT-09), Wiener phase noise with keyed increments, and a front end with full-scale mapping, gain and optional clipping.

D. What the release does not model

Table VI lists what the current release does not model or provide, and what it offers instead; Section VIII-C names the four limits that matter most for reading the results of this paper.

TABLE VI
WHAT THE CURRENT RELEASE OF `SREMU` DOES NOT MODEL OR PROVIDE, AND WHAT IT OFFERS INSTEAD.

Not modelled or not provided	Instead
Unmodified UHD applications; a virtual USRP driver	UHD-shaped API and sockets; virtual-USRP emulation is the approach of ACHEM [33]
Hard real time; paced timing on Windows	Soft real time with counted misses; timing results from Linux hosts
Ray tracing, buildings, terrain profiles, several knife edges on one path, walls of order two, RIS–wall paths	Explicit scenes: LOS, flat ground, order-one walls, one knife edge per path, additive diffuse term
RIS–RIS paths; polarization	Single-bounce RIS paths; scalar co-polar fields
Attenuation of RIS hops by occluders; fading or multipath on RIS hops	Occluders act on the LOS path and its ground and wall images; RIS hops are deterministic free-space paths (optional flat-ground images)
Mutual coupling between differing neighbour states	Per-state tables, which hold coupling for identical neighbours (locally periodic) [6]; a coupling-hook protocol, implemented as no coupling
Angle-dependent tables outside one plane of incidence	Single-plane tables; off-plane components projected out, or refused on request
Switching, transient or timing behaviour compared with measurements	Transition profiles as an engineering choice; command records with requested and applied samples
Frequency-dependent Γ filters; non-reciprocal tables	Signals wider than a table's validity bandwidth and non-reciprocal tables are rejected
Sampling-frequency offset, IQ imbalance, amplifier and converter models	CFO, phase noise, full-scale clipping, `sc16` saturation
Rate conversion between overlapping bands; several channels per stream	Linked streams share one rate; one channel per stream
Web server, dashboard, authentication	Static HTML report; loopback sockets by default
E2SM-RIS adapter; conformance to a standard; GPU IQ path; vectorized environments	Future work (Section IX)
Learning-based controller research; IQ backends of `RisEnv`	Controller protocol; `RisEnv` on the batch path; a UCB1 baseline

APPENDIX D
VERIFICATION AND VALIDATION DETAILS

This appendix extends Section V.

A. Test record

Table VII summarizes the test record per tier, with the identifiers that passed and the number of implementation tests. No skipped test carries an identifier: the skipped tests check fallback behaviour for modules that are now present, and a processor-affinity call that this platform lacks. An identifier counts as passed when at least one of its implementation tests passed and none failed or was skipped. Tests of the rendering path count once per backend. Appendix I lists the commit and date of this record with those of every other result; the record does not state its host.

The analytic tier spans the whole emulator. On the physics side it checks the Friis equation, the equality of K1 with three published forms, image theory, the N^2 law of a focused panel, the two-ray model, the knife edge and the losses of finite alphabets. On the runtime side it checks exact sample counts at incommensurate rates, the switch sample of a distant receiver, late and tied commands, barrier determinism and bitwise invariance under any chunking of the client calls. Section III cites these identifiers where it states a property.

B. MATLAB code-to-code references

G1 sums K1 per element with toolbox primitives for laboratory-like layouts and for the geometries of Tang et al. G2 and G3 compare steered and one-bit beam patterns with MATLAB's RIS reflector object. G4, G5 and G6 compare the frequency offset, thermal noise and fading statistics with communications-toolbox objects. G7 evaluates the far-field closed form of ETSI GR RIS 003 [6], and G8 is MATLAB's two-ray channel. MATLAB is not needed to run them: the outputs are stored with the script hash and the MATLAB version. MATLAB's reflector uses unit-gain cells, so G2 and G3 compare beam shapes after a documented constant is removed. They say nothing about absolute RIS power.

TABLE VII
VERIFICATION SUMMARY FROM THE TEST RECORD OF EMULATOR COMMIT 1e1fcab (APPENDIX H). AN IDENTIFIER PASSES WHEN AT LEAST ONE IMPLEMENTATION TEST THAT CARRIES IT PASSED AND NONE FAILED OR WAS SKIPPED; RENDERING-PATH IDENTIFIERS COUNT ONCE PER BACKEND. TABLES XV AND XVI GIVE THE PROPERTY, ORACLE AND TOLERANCE OF EVERY IDENTIFIER. PH-12, A REGRESSION VALUE OF THE EXPERIMENTAL P.526 MODULE THAT LIES OUTSIDE EVERY GATE, IS NOT COUNTED; ITS IMPLEMENTATION TESTS RUN IN THE EXPERIMENTAL TEST TIER, OUTSIDE THE CONTINUOUS-INTEGRATION AND NIGHTLY RUNS THAT THIS RECORD COVERS.

Tier	What is compared	Identifiers passed	Passed	Tests
PH	Physics closed forms and exact properties: Friis, area-tied kernel equals published forms, aperture refinement, image theory, focusing, ground, knife edge, alphabets and optimum, Doppler, interpolation, fractional delay, noise, linearity	PH-01–PH-11, PH-13–PH-32	31/31	352
RT	Runtime: exact counts at several rates, round trips, bitwise invariance, switch samples, late commands, ties, barriers, oscillator continuity, transitions, replay, paced counters, backends	RT-01–RT-19	19/19	192
IO	Interfaces: malformed frames and messages, two-process bitwise equality, socket commands, caps and loopback, SigMF, status codes, wrappers, raw mode	IO-01–IO-08	8/8	130
ST	Statistical: random-waypoint law, diffuse autocorrelation and level crossings, quantization and failure Monte Carlo, BER vs theory, CFO vs CRB	ST-01–ST-05	5/5	30
G	MATLAB R2026a code-to-code: per-element kernel, beam patterns, CFO, thermal noise, fading statistics, ETSI far field, two-ray channel	G1–G8	8/8	30
VA	Consistency of engine paths: served IQ path vs batch evaluation (E-A), exhaustive optimum vs every controller, IQ path equals batch path, sample-exact switch	VA-1, VA-3–VA-5	4/4	20

TABLE VIII
EMULATOR AGAINST THE DIGITIZED CHAMBER MEASUREMENTS OF TANG ET AL. [36] PER GEOMETRY (TRANSMITTER DISTANCE d_1 AND ANGLE θ; NINE RECEIVER DISTANCES EACH): MEASURED MINUS EMULATOR IN DB, AND THE LARGEST $|\text{EMULATOR} - \text{PUBLISHED MODEL CURVE}|$. MEASURED POINTS AND CURVES: DIG; EMULATOR: SIM-SREMU; LINK BUDGET: REPORTED.

d_1 (m), θ (°)	n	Bias	SD	RMSE	max$\lvert e\rvert$	Curve
1, 10	9	0.79	0.37	0.86	1.48	0.012
2, 10	9	1.15	0.39	1.21	1.59	0.013
1, 45	9	1.06	0.29	1.09	1.48	0.013
2, 45	9	0.91	0.42	0.99	1.40	0.012
all	36	0.98	0.38	1.05	1.59	0.013

C. Aperture refinement and Tang et al. per geometry

The far case of Fig. 2(a) is specular reflection; the near case is an anomalous angle, whose continuous physical-optics integral we evaluate independently. There, one point per cell is off by 30 dB for the coarsest cells and converges slowly. At the smallest count with a pitch below half a wavelength (1024 cells) it is still off by 0.5 dB. This is the model-form difference of Section IV-C, not a refinement error, and it is why the default switches to patches for larger cells.

Table VIII breaks the comparison of Section V-C down by geometry: the digitized measurements of Tang et al. (DIG) against the emulator with the reported link budget (REPORTED). The largest difference is 1.59 dB and the per-geometry means span 0.79–1.15 dB; the published model, too, lies below every measured point. Since K1 equals Tang et al.'s model for one point per cell, the table characterizes that model class against their chamber data; it does not test reconfiguration, steering or timing. The extraction reads the plotted vertices and marker centres of Fig. 10 of the arXiv version 2101.08607v2 of [36], so its error is that of the axis calibration, which the curve agreement bounds; the distances carry a constant calibration offset of a few millimetres, removed by snapping them to the nominal grid. The source states no measurement uncertainty, and the points of one geometry are not independent. We do not fit away the constant part of the error. It may come from the reported link-budget terms or from the reported amplitude of the coding state, $|\Gamma| = 0.9$: a unit amplitude would raise the model by 0.92 dB [36].

D. Laboratory consistency check: protocol and outcome

The map was measured at IIT Bhilai with two Ettus B200 radios and vertical whip antennas in front of six binary tiles fabricated by the RIS group there, whose RIS hardware includes the switched unit cells of [83]. The map and the two state phases of Section V-D appeared in an unpublished ANTS 2025 conference submission by the author and colleagues, which drew a qualitative validation claim from the map; this study digitized that colour map, the form in which the measurement was available to it, and its pre-specified null result supersedes that claim. The geometry of the measurement was not recorded to the precision a model needs: the distance from the transmitter to the tiles, the angular span of the tiles, the cell type (whether the tiles use the cell of [83] was not recorded, so the prior covers it and a square half-wavelength cell) and the measurement positions are known within ranges. The five models of the plan are line of sight alone (M0), a log-distance fit that also fits its exponent (M0n), a uniform mirror (M1), the laboratory configuration (M2) and M2 with a floor (M3). The offsets are fitted on three rings (18 cells) and scored on the other two (12 cells). C1 (level consistent) needs a median absolute bias of at most 1.5 dB and a median RMSE of M2 at most 0.5 dB above that of M0n. Because each offset absorbs the unknown power and gains, C1 tests the level of the held-out rings, not absolute power. C2 (the RIS model explains spatial structure) needs a median RMSE gain of M2 over M0 of at least 1.0 dB, a paired-bootstrap interval that excludes zero in at least 80 % of the draws, and a Spearman correlation of at least 0.3 whose interval excludes

TABLE IX
LABORATORY CONSISTENCY CHECK: TEST-RING METRICS OVER THE PRIOR DRAWS OF THE PRE-SPECIFIED PROTOCOL (MEDIAN, WITH 5–95 % RANGE FOR THE RMSE). ERROR = MEASURED − PREDICTED. MEASURED: DIG; PREDICTIONS: SIM-SREMU; GEOMETRY AND $|\Gamma|$: ASSUMED (PRIOR).

Model	RMSE (dB)	Bias (dB)	MAE (dB)	max$\|e\|$ (dB)
M0 (line of sight)	3.96 (3.92–3.99)	−0.59	2.18	12.6
M0n (log-distance)	3.77 (3.73–3.81)	−0.47	2.22	11.9
M1 (mirror)	5.00 (3.36–6.75)	−0.64	3.50	13.5
M2 (lab configuration)	5.11 (3.27–6.66)	−0.60	3.53	13.9
M3 (M2 + floor)	5.18 (3.30–6.79)	−0.63	3.60	14.1

Decisions: C1 fails; C2 fails; C3 fails; decoy C2 detection rate 0.00 (underpowered).

zero. C3 is C2 with M1 in place of M0. The plan followed a descriptive inspection of the map: the cell grid, the per-cell digitization uncertainty, the two cells pinned to the colour scale, the mean and spread of each ring and a log-distance fit to all cells were known when it was written, but no RIS model had been fitted to the data. The hash was committed to the emulator repository (commit 101ee87) and pushed on 11 Oct 2026, 00:22:55 UTC; GitHub's server-side push log, exported later, records that push (Appendix I). Two commits followed: an option of the run script that names the columns of the measured file, and a test change (commit 7a63287), pushed at 00:30:30 UTC; the unblinded run started at 00:30:44 UTC from a clean tree of that commit, and its result (commit 2333b60) was pushed at 00:31:40 UTC. No public registry deposit (Zenodo or OSF) was made before unblinding; the plan, its hash and the lock records can still be deposited, as a retrospective record.

In the decoy power analysis, C1 held in 0 and C3 in 0 of the 500 synthetic maps drawn from M2.

Table IX and Fig. 7 give the outcome. The 5–95 % range of the median test RMSE of M2 over the prior draws is 3.27–6.66 dB, with a median M2 bias of -0.60 dB. The bootstrap interval of the RMSE gain of M2 over M0 excludes zero in 0 % of the draws, and the 95 % interval of the Spearman correlation is -0.76 to 0.34. For orientation, outside the decision rules: the observed RMSE gain lies close to the median of the decoys drawn from M2 (-1.16 dB) and above that of decoys drawn from M0 (-2.13 dB). The digitized map does not discriminate the RIS model from LOS-only at the available geometric knowledge. One cell, pinned to the edge of the colour scale, has a residual of -13.7 dB.

In the diagnostic, none of 8 single facts (for example the transmitter-to-tile distance to a few millimetres, the tile angles, the cell type, the antenna pattern or the state reflection coefficients, or even the measurement positions alone to ± 1 mm) lifted the C2 detection rate above zero. Knowing the as-built tile geometry exactly and the measurement positions to ± 1 mm gave 0.53 (95 % interval 0.46–0.60, 200 decoys); with positions to ± 5 mm it stayed at zero, and knowing every geometric quantity exactly gave 0.63. A geometry recorded to about a millimetre needs, for example, a fixture or a laser tracker. Even then the detection rate stays close to the pre-set threshold.

E. Frequency offset, Cramér–Rao bound and bit error rate

A receiver oscillator offset f rotates the received baseband at f. The sweep used a BPSK burst of 2389 samples at 5880 S/s; besides the least-squares phase estimator, a lag-1 estimator recovers f within 1.9×10^{-8} Hz (Fig. 8(a)). The sweep, with its offset grid and its BPSK burst, appeared in the unpublished ANTS 2025 submission (Section II); the emulator regenerates it here.

Fig. 8(a) also shows the generic pitfall. At 5880 S/s a tick of 2 ms holds 11.76 samples. A time base that delivers a whole number of samples per tick, here 11, drops samples without notice, and every frequency estimate is scaled by the ratio of the two counts: a slope of 1.0691, an error of 6.9 %. We reproduced the effect by re-timing the emulator's own output. Exact rational sample counts (Section III-B) remove it.

The noise runs used per-sample SNRs of 0, 10 and 20 dB and 200 keyed trials at each of 7 offsets, and the periodogram estimator works on 2000 samples. Its pooled RMSE is 1.00, 0.99 and 0.96 times the bound. An unbiased estimator cannot beat the bound, so values below one are sampling error of the pooled estimate. The lowest lies 2.3 standard errors below one. We checked that the emulator's noise is white and has the stated power, and the gate does not depend on this value. The worst single offset at the two gated SNRs is 1.06 times the bound, within the pre-set limit of 1.5 (Fig. 8(b); V-CFO-4, ST-05). The lag-1 estimator follows its own small-error theory but not the bound: its variance falls with the number of samples, the bound's with its cube.

In the study E-F, Fig. 8(c), BPSK and QPSK pass through a static lab-like link with the emulator's physical noise. The transmit power sets E_b/N_0 from 0 to 10 dB, and each point collects at least 200 errors or 2×10^7 bits. With a least-squares channel estimate from 64 pilots, theory falls outside the interval at 5 points, all at $E_b/N_0 \leq 6$ dB, as expected from the estimation loss. ST-04, a reduced version in the test record, repeats the genie check with 99 % intervals.

F. Doppler and interpolation

Fig. 9(a) shows the Doppler run of Section V-F, in which the receiver approaches the transmitter, over one second. The RIS contribution is taken as the difference between runs with and without the panel. Near a null of the panel pattern the phase of the sum jumps and no frequency is defined. We therefore compare the samples where the reference RIS amplitude exceeds 10 % of its maximum (88 % of the samples). There the largest deviation is 0.33 Hz and the median 0.04 Hz. The interpolation bound applies to the whole channel, so a weak RIS term inherits a larger relative error.

Fig. 9(b) shows the error of the snapshot grid. At the automatic period of 741 μs the largest relative error against a channel evaluated at every sample is 7×10^{-6}. The error grows as T_u^2 and reaches 7×10^{-4} at 7.4 ms. The engine's own estimate, `verify_interp`, follows it but reads low, by up to 40 % at the largest period. PH-18 checks the bound against per-element brute force in three scenes on both backends.

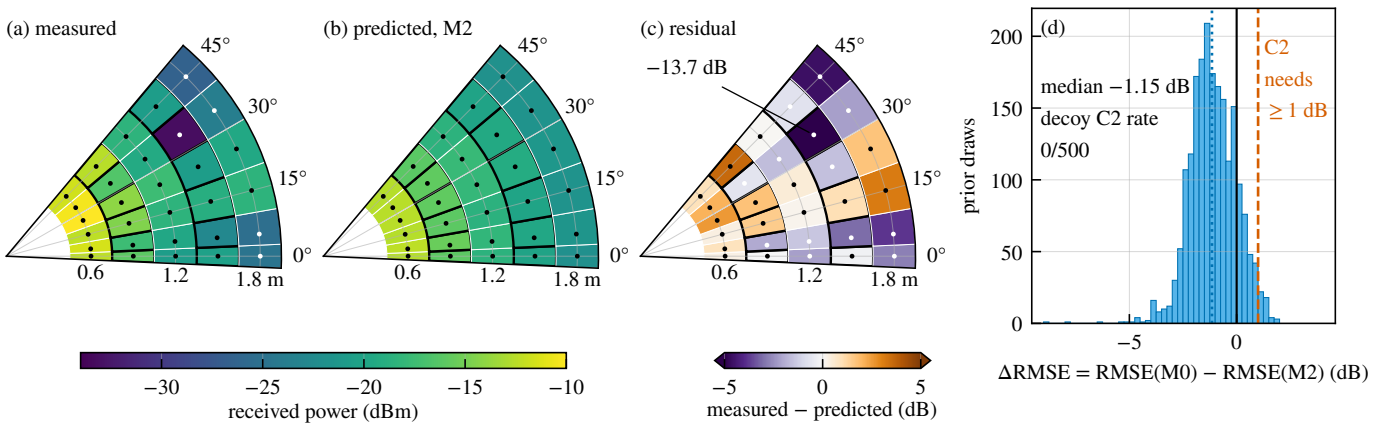


Fig. 7. Laboratory consistency check under geometric uncertainty. (a) Digitized map of a laboratory measurement collected at IIT Bhilai. (b) Median prediction of the laboratory-configuration model M2 over the prior draws. (c) Measured minus predicted; the colour scale is clipped and the outlier is marked. Test-ring cells are outlined. (d) RMSE gain of M2 over the line-of-sight model M0 on the test rings over the prior draws, with the pre-specified C2 threshold. Data classes: DIG (measured), SIM-sremu (predictions), ASSUMED (geometry and reflection-coefficient prior, decision rules).

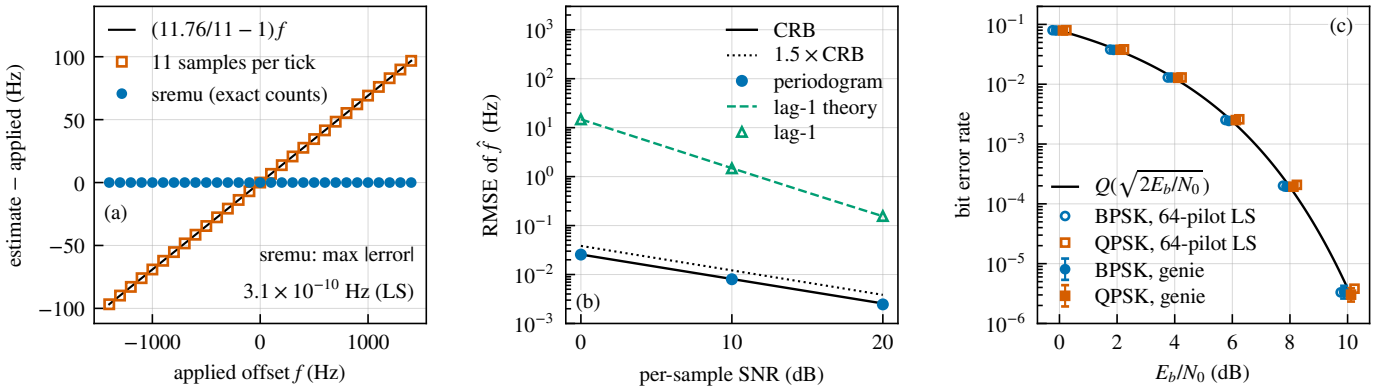


Fig. 8. (a) Error of the estimated frequency offset versus the applied offset, noise off: the emulator (exact sample counts) and a generic pitfall, the emulator's own output re-timed with a whole number of samples per tick, with its analytic slope. (b) RMSE of two estimators versus SNR against the Cramér–Rao bound and the lag-1 small-error theory. (c) BER versus E_b/N_0 through the emulator with genie and pilot-based detection, Clopper–Pearson intervals, and theory; markers are offset horizontally for legibility. Data classes: SIM-sremu (emulator runs), ANALYTIC (bound, theory, slope), ASSUMED (unit amplitudes of the lab states in (c)).

APPENDIX E
BENCHMARK PROTOCOL AND FURTHER RESULTS

This appendix extends Section VI.

A. Hosts and protocol

Table X lists the hosts and the emulator commit of each run; the cross-tool context (B6) comes from a laptop. The lockstep scaling runs B1 and B2 predate the pinned code commit, at which B2-F45, B3, B4 and B4-classic ran; B1 and B2 recorded a dirty flag, although their uploaded file tree equals their commit's (Appendix I). B2 and B2-F45 packed up to 11 points onto the server at a time, each pinned to 16 logical CPUs of whole cores in one NUMA node. A B3 point runs up to 3 repetitions of 60 s, each with at least 46 876 pages, and counts as Sustainable when every repetition is; the link count of a series is doubled while it is Sustainable, and an unsustainable repetition ends the point and the series. Headless runs have no external inputs, so their late-input condition holds trivially. The benchmark scenes have one carrier and a non-zero surface path on every link; the rest of each host was idle apart from the B4 command client, and the laptop of B6 was not idle. In the benchmark scenes, 8 links are, for example, 2 transmitters and 4 receivers. Because a page is computed from its release on, every page misses a deadline of zero slack. Another host needs its own runs.

B. Lockstep points, surfaces and threads

Fig. 10 shows the lockstep scaling of Section VI: the RTF over links and element counts, static and with mobility (a); the cumulative feature sets at one link (b), with RTF 6.95 without noise (F0), 5.14 with noise (F1), 4.17 with oscillator impairments (F2), 1.56 with mobility (F3) and 1.46 with recording (F4); the aggregate throughput, which peaks with noise (F1) at 32 MS/s with 2 receivers at 23.04 MS/s and an RTF below one (c); the surface count (d) and the thread count (e). One link with one surface keeps up at 23.04 MS/s (RTF 1.16), and with two surfaces (RTF 1.01) but not with four (0.94). Thread counts above the slot size ran on slots of as many logical CPUs. At 8 and 64 links the best count, 8 threads, gives 1.4 and 1.9 times the one-thread RTF, and every larger count was slower.

TABLE X
PLATFORMS AND SOFTWARE OF THE BENCHMARK RUNS (MEAS). EMULATOR COMMITS: B1 AND B2 AT `309c633` (DIRTY FLAG RECORDED, UPLOADED FILE TREE EQUAL TO THE COMMIT'S); B2-F45, B3, B4 AND B4-CLASSIC AT `BF08268` (CLEAN TREE); B6 AT `06CDF7A` (CLEAN TREE). SERVER: B1 ON 16 VCPUS WITH THE REST OF THE HOST IDLE, B2 AND B2-F45 AS UP TO 11 CONCURRENT POINTS OF 16 VCPUS; PACED HOST: ENGINE ON 16 VCPUS, B4 COMMAND CLIENT ON 4, THE REST IDLE. B3, B4 AND B4-CLASSIC RAN ON THREE INSTANCES OF ONE TYPE, B2 AND B2-F45 ON TWO OF ANOTHER. N/R: NOT RECORDED BY THE RUN. NO NUMBER IS COMPARED ACROSS INSTANCE TYPES OR WITH THE LAPTOP.

Runs	Host	CPU	Cores/threads	RAM (GiB)	OS	Python	NumPy	Numba	pyzmq	MATLAB
B1, B2, B2-F45	AWS c7i.48xlarge	Xeon Platinum 8488C	96/192	372	Ubuntu 24.04, kernel 7.0	3.12.3	2.5.3	0.68.0	27.2.0	—
B3, B4, B4-classic	AWS c7i.8xlarge	Xeon Platinum 8488C	16/32	62	Ubuntu 24.04, kernel 7.0	3.12.3	2.5.3	0.68.0	27.2.0	—
B6	Laptop	Ryzen 9 8945H	8/16	n/r	Windows 11, Balanced	3.12.4	2.5.3	0.68.0	27.2.0	R2026a

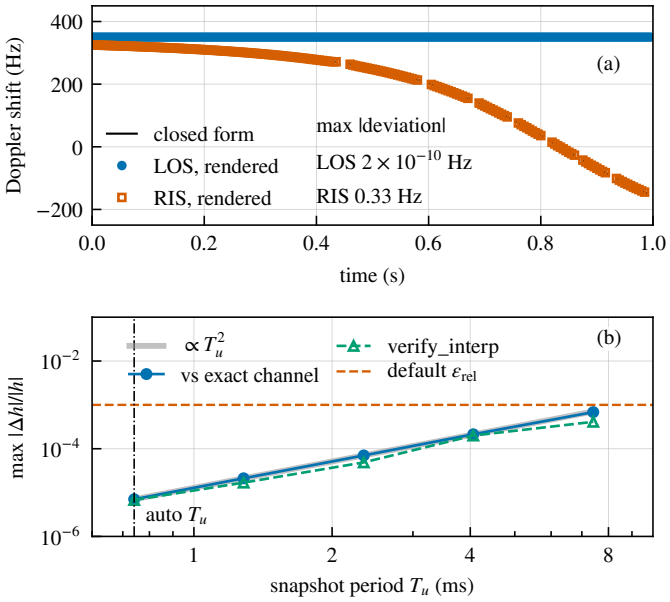


Fig. 9. (a) Doppler shift of the rendered line-of-sight path and RIS contribution of a moving receiver against the closed forms; RIS samples near nulls of the panel pattern are omitted. (b) Largest relative interpolation error over one second versus the snapshot period: against a channel evaluated at every sample, and as estimated by the engine. Data classes: engine runs and closed forms with the target (SIM-sremu; ANALYTIC).

Of 909 planned B2 points, 889 ran; the other 20 were not started because a smaller point of their series had fallen below the pilot floor or failed. Of the points that ran, 876 completed, 9 of the heaviest mobility points fell below the pilot floor (a pilot estimate, no measurement), and 4 failed. Every failure was a point with the native ZMQ data plane (F5): three receive time-outs came from an error of the benchmark, whose served run ended before the clients' streams, and the fourth, also a benchmark error, came from summarizing a point whose pilot was below the floor. Both were fixed in the benchmark, not in the emulator, and F5 was rerun (B2-F45).

Concurrent points share the last-level cache, memory bandwidth and the power budget of the host. Rerun alone, the 6 reference points of the isolation check were up to 16 % faster (ratio 0.99–1.16), so the packed values in Fig. 10 are conservative. The link limit lies within this margin: at 16 links the packed RTF is 0.90–0.94 over N, while the same base configuration measured alone in B1 reached 1.07. We therefore state the limit as 8 links packed and between 8 and 16 links alone.

C. Data plane

The F5 figure is end to end: the virtual time delivered to the receiving client process divided by the wall time from the served engine's warm-up boundary to the client's last sample, so a client cannot shorten it. The 100 stress runs of the base point reached RTF 0.79–0.90 (median 0.86) with no stall and no bad packet. In 30 runs whose receiving clients paused 3 s before their warm-up completed, it fell to 0.10 (median), while the client's own read rate, a figure of the client alone that such a pause inflates, reached 56. The single isolation reference point of B2-F45 ran 1.17 times faster alone. In the E-F measurement (lockstep, on a busy c7i.48xlarge), through the UHD-shaped API a buffer of 1000 samples took 21 μs to send in process and 1755 μs to send and receive across two processes (medians; p99 38 and 6783 μs), and streaming over native ZMQ sustained 1.50 MS/s; receiving a buffer in process, which includes computing it, took 295 μs (median), alternating send and receive sustained 2.72 MS/s, and the ZMQ rate was measured with buffers of 10 000 samples. Other work of the regeneration job kept the host up to 37 % busy during these measurements.

D. Paced series and short pages

The 8 links with 1024 μs pages missed at most a fraction 5.1×10^{-5} of the deadlines at 1 ms slack in a repetition. The single link at 23.04 MS/s uses pages of 711 μs (p99 672 μs, miss rate 3.6×10^{-5}); at 5.76 MS/s the pages last 356 μs and at 0.1 MS/s 1.28 ms. With pages of 512 μs the 4 links have a worst p99 of 632 μs; four surfaces sustain as many links (4), and $N = 4096$ one.

Two mechanisms end the short-page series. At 512 μs, 8 links keep the engine busy for a median 0.71 of each page and miss 2.2 % of the deadlines: computation. The 256 μs series and the $N = 4096$ series instead passed two repetitions of their last point and failed the third, through stalls of up to 11 ms (miss rate 1.28×10^{-3}) and 142 ms; the runs record no stage breakdown that would identify the stalls. With mobility (F3), one link misses 17 % of the deadlines at 512 μs pages in the single repetition run: the geometry snapshot is recomputed every 2.25 ms, 23 % of the pages hold a recomputation, and such a page takes up to 1.3 ms (p99), although the mean load stays below the wall clock. Not tried: mobility at other page

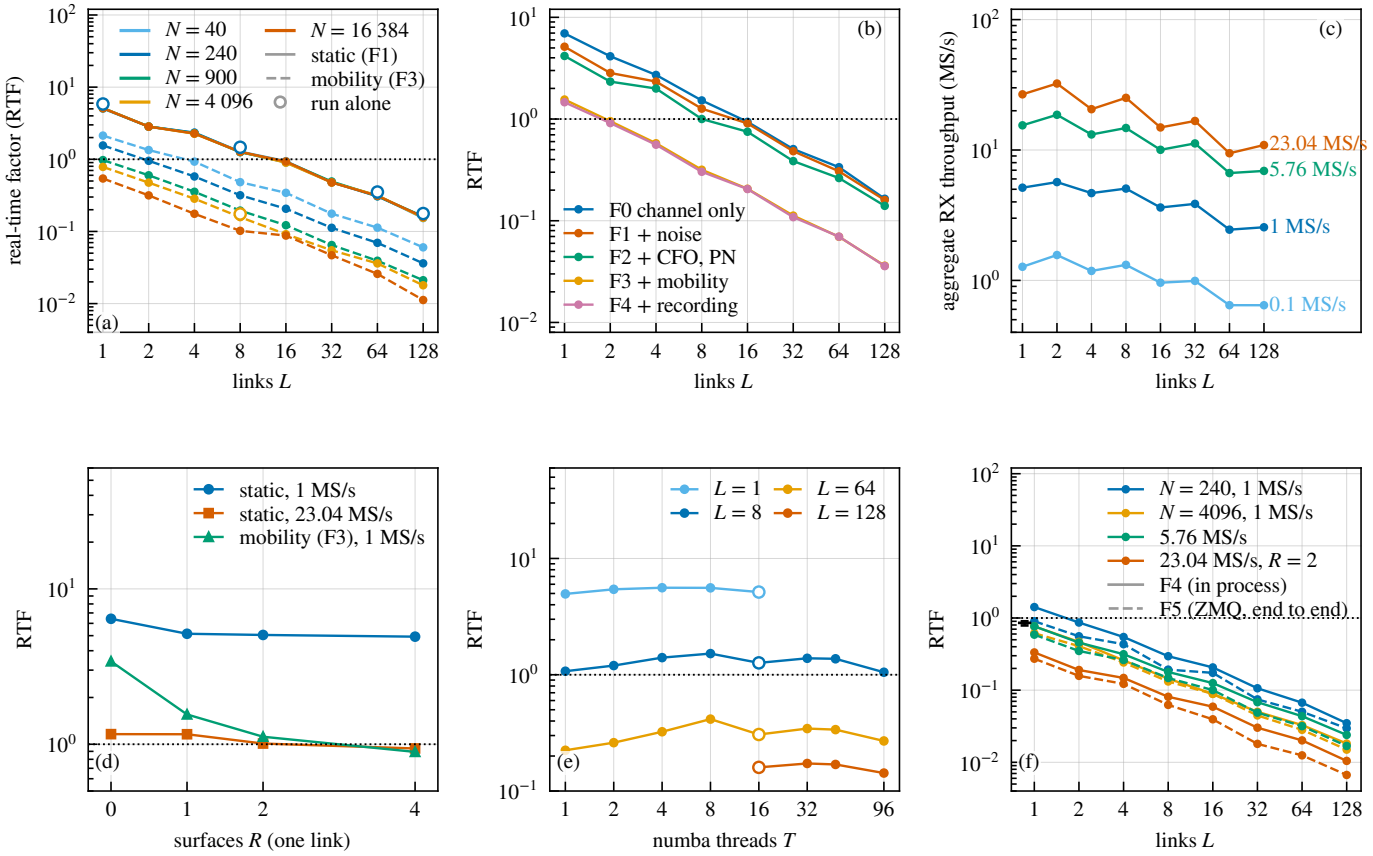


Fig. 10. Lockstep scaling on AWS c7i.48xlarge instances (MEAS). (a)–(e): B2, up to 11 points running concurrently. (a) RTF versus the number of links L at 1 MS/s with one surface, for every element count N, static (F1, solid) and with mobility (F3, dashed); open markers: isolation reference points run alone; dotted: RTF = 1. (b) RTF versus L for the cumulative feature sets F0–F4 at $N = 240$. (c) Aggregate received throughput, $n_{\mathrm{rx}} f_s$ RTF with n_{rx} receivers among the L links, for four sample rates. (d) One-link RTF versus the number of surfaces R ($N = 240$). (e) RTF versus the numba thread count ($N = 240$, one surface, 1 MS/s, F1; open markers: the default grid points). (f) B2-F45, another instance at another commit: RTF of the native ZMQ data plane to a transmitting and a receiving client process (F5, end to end, dashed) and of the same scenes in process (F4, solid) in the same run; bar: range and median of the 100 stress runs of the F5 base point.

durations or link counts, more than one surface with mobility, and paced runs with client processes on the data plane.

E. Control loop

At the highest lockstep rate the last command of a run switches after the end of the run and is never applied; the check counts such commands as scheduled after the end, not as failures. The per-command engine cost of Section VI-C (barrier, decision and configuration change) holds at 100 Hz and above, and is computed from the RTF with and without the loop. It includes splitting the page at the switch and rebuilding the link statics; the laptop run B6 times a configuration change with the geometry cached at 77–109 μs, and these runs record no stage breakdown that would apportion the rest.

Served paced engines render small pages serially, let idle OpenMP workers sleep and exempt their objects from garbage collection during the loop. A served repetition is *backlogged* when its median page latency exceeds one page or more than 1 % of its pages miss a slack of 5 ms; a backlogged repetition's command latencies measure the backlog, so its late and exact fractions are not counted. In the served-1ms plan, timed commands carried leads of 2–12 ms. A timed command is late when the engine takes it after its switch sample was computed. At its receipt the switch lies a margin ahead of the pacer's schedule: the lead minus the delay to the server and the error of the client's estimate of virtual time. The p1 of this margin per repetition was 0.5–1.4 ms with a 2 ms lead, where the late fraction of every repetition exceeded the share of commands below its p1, and 2.9–3.3 ms with 4 ms, where it fell short of that share in every repetition. A single margin threshold would thus lie between about 1.4 and 2.9 ms, of the order of one page plus its computation; the runs do not record when the engine took each command, so this is an inference. Pooled over the repetitions, late fractions were at most 0.17 % (4 ms; 0.34 % in one repetition) and 0.06 % (6–12 ms). The client's round trip of a command reached a p99 of up to 6.1 ms; the apply delay (Fig. 3(d)), which excludes the lead, had medians up to 2.1 ms and a p99 up to 4.7 ms; its largest value, 130 ms, came with a page stall of about that length in one repetition, and asap commands had rare visible-latency outliers of up to 17 ms. The latency floor is consistent with the engine's work on the page with the configuration change; that the spread comes from the hand-over between the control-socket thread and the paced loop, which share one Python interpreter, is a hypothesis. Page deadlines are missed

between commands too: at 10 commands/s the miss rate at 5 ms slack was at most 2.4×10^{-4}. At 1000 commands/s page latencies grew to 4.6 s.

The original plan (B4-classic, another instance) gives the same picture with pages of 512 μs and a lead of 2 ms: 5.7–20.1 % of the pages missed the 1 ms deadline at 10 and 100 commands/s, against at most 8.5×10^{-6} for the same links and pages headless (B3), so the served baseline is not marginal headless; 1.6–4.9 % of the timed commands were late, the asap visible-latency p99 stayed at or below 5.1 ms, and from 1000 commands/s on every repetition was backlogged, with 281–446 commands/s achieved.

F. Cost model

B1 fits a page cost model, a constant plus linear terms for snapshots, fractional-delay filtering, receivers and paths, to 20 engine points measured on an otherwise idle server. The fit is indicative: its relative RMS error is 39 % and its largest error 69 %. The constant term, 65 μs per page, is 76 % of the single-link page cost of 86 μs, so short pages pay mostly fixed cost. The largest errors are underestimates of mobility points. There, a snapshot costs about 0.61 ms plus 0.50 ms per link, with little dependence on the number of surface rows and quadrature points RQ at the sizes of these points; a diagnostic fit with these terms reaches 12 %, but with 8 coefficients for 20 points it is not used for prediction, and at the larger element counts of B2 the snapshot cost does grow with N (Fig. 10(a)). The kernel itself scales as expected: the snapshot time grows as $(RQ)^{1.06}$ on the NumPy reference backend for RQ from 40 to 1.0×10^6, and as $(RQ)^{0.76}$ with numba up to 1.7×10^7, where fixed costs dominate the small sizes.

G. Cross-tool context

For context, B6 times one evaluation of the same near-field surface sum on the laptop for $N = 256$ to 4096 elements with `sremu` and with MATLAB R2026a; where both compute the same model the results agree to 7.8×10^{-14} dB. `sremu` takes 307–466 μs for new positions and 77–109 μs for a configuration change with the geometry cached; a vectorized MATLAB script takes 92–284 μs, a per-element MATLAB loop 84–1903 μs, and the toolbox's far-field `phased.RISReflector`, a different model, 251–539 μs. These numbers are not a ranking: the calls do different work (the `sremu` times include Python dispatch and the line-of-sight path; the MATLAB times exclude its start-up), they come from one machine whose other processes were busy for 11 % of the time during the `sremu` timings, and a streaming emulator's cost is per page, not per call.

APPENDIX F
CASE-STUDY DETAILS

This appendix extends Section VII.

A. E-B: setup, cases and lab-like geometry

Three lessons stand out in E-B: an equalizer trained before a switch inside a packet sees an error as large as the signal after it; a long linear transition spreads the damage but does not remove it; and with a strong line of sight QPSK tolerates the wrong configuration, so the timing of a command matters where the RIS path dominates or the margin is small.

In the arc geometry each uncertain distance and angle takes the middle of its known range, the two receivers stand at different angles at positions that are an engineering choice (ENG), and E_s/N_0 at RX1 is set against the physical noise. A lab-like SIM geometry with one panel of three tiles is a supporting case. Three assumptions apply to both geometries. The two states of the laboratory cells have unit amplitude ($|\Gamma| = 1$, ASSUMED), because their magnitudes were not measured. During a transition the response follows the complex-linear path between the states, an engineering choice (ENG). When the line of sight is blocked, an opaque plate removes the direct paths, and the RIS hops stay unattenuated because occluders act on the line of sight alone. On the arc the plate crosses 152 of the 240 segments from the transmitter to a cell and none from a cell to a receiver, so a physical plate would shadow most of the incident illumination: the blocked results, PERs and EVMs alike, belong to this idealized regime, and the cases with the line of sight present are the physically consistent ones. The packets run at 4 samples per symbol and 1 MS/s.

The arc run fixed its whole E-B2 grid before any point ran (Table XI): guards of 0.1, 0.5 and 1 ms; five jitters; timed commands with three leads, asap and tick-based commands; the laboratory alphabet with the line of sight present at $E_s/N_0 = 25$ dB and at 15 dB (a sensitivity) and blocked at 25 dB; and the ideal 1-bit alphabet with the line of sight present at both E_s/N_0. On the arc each controller message carries one command per tile. With the line of sight present at 25 dB QPSK tolerates the wrong configuration, although the EVM of the affected packets rises. With a lead of 0.5 ms and the 0.5 ms guard, timed commands fail already at $J = 1$ ms (15 % with the line of sight present at 15 dB and 21 % with it blocked); with 5 ms they lose no packet at any jitter.

The closed form of Section VII-A follows from the timeline. With $\Lambda = \Lambda_0 + JU$, $P(\Lambda > x) = \min\{1, \max[0, (\Lambda_0 + J - x)/J]\}$, or $\mathbf{1}[\Lambda_0 > x]$ for $J = 0$. A packet starts T_g after an asap command is issued and $T_\mathrm{g}/2$ after the requested time of a timed one, which a late timed command misses by $\Lambda - \ell$; the switch thus lands after the preamble, of $T_\mathrm{pre} = 0.256$ ms, with probability $P(\Lambda > T_\mathrm{g} + T_\mathrm{pre})$ or $P(\Lambda > \ell + T_\mathrm{g}/2 + T_\mathrm{pre})$. A tick-based command waits a further W for the next tick of period $T_\mathrm{t} = 2$ ms. Over a run the guard starts visit every multiple of 4 μs (8 μs at the longest guard) of the tick phase equally often, so W is uniform on $(0, T_\mathrm{t}]$ and the probability is $P(\Lambda + W > T_\mathrm{g} + T_\mathrm{pre})$, the integral over U of a piecewise-linear function, which is $1 - (T_\mathrm{g} + T_\mathrm{pre} - \Lambda_0)/T_\mathrm{t}$ for $J = 0$. With the line of sight blocked, every asap and timed cell with a zero closed form lost no packet; of the 14 with a positive one, 10 contain it in their interval and none lies below it beyond its interval. The measured PER exceeds it by up to 8.8 points, most at the 0.1 ms guard, where many switches fall in the last part of the preamble and corrupt the channel estimate; switches up to 200 μs into the preamble cost nothing. Tick-based cells exceed their closed form by at most 1.5 points, except the 5 cells with $J \geq 1$ ms in which a later command overwrote

TABLE XI

E-B2 ON THE ARC, THE FULL PRE-DECLARED GRID: PACKET ERROR RATE IN PERCENT OF 2000 PACKETS PER CELL, BY SEMANTICS (TIMED: LEAD ℓ), GUARD T_{g} AND JITTER J OF THE LATENCY $\Lambda = 0.2\,\mathrm{ms} + JU$. CF: CLOSED-FORM PROBABILITY THAT THE SWITCH LANDS AFTER THE PREAMBLE (APPENDIX F); B: LABORATORY ALPHABET, LINE OF SIGHT BLOCKED, $E_s/N_0 = 25\,\mathrm{dB}$; L, I: LABORATORY AND IDEAL 1-BIT ALPHABETS, LINE OF SIGHT PRESENT, 15 dB. WITH THE LINE OF SIGHT PRESENT AT 25 dB NO CELL OF EITHER ALPHABET EXCEEDS 3.1 %. †: TICK-BASED COMMANDS OVERWRITTEN BEFORE THEIR TICK, WHICH THE CLOSED FORM IGNORES. CLOPPER–PEARSON 95 % HALF-WIDTHS ≤ 2.2 POINTS (UPPER LIMIT 0.18 % AT ZERO ERRORS). SIM-SREMU; LABORATORY AMPLITUDES ASSUMED.

		$J = 0$				$J = 0.1\,\mathrm{ms}$				$J = 0.3\,\mathrm{ms}$				$J = 1\,\mathrm{ms}$				$J = 3\,\mathrm{ms}$			
Semantics	T_{g} (ms)	CF	B	L	I	CF	B	L	I	CF	B	L	I	CF	B	L	I	CF	B	L	I
timed, $\ell = 0.5\,\mathrm{ms}$	0.1	0	0	0	0.05	0	0	0	0.05	0	0	0	0.05	39.4	40.8	34.6	31.4	79.8	81.6	79.2	78.3
	0.5	0	0	0	0	0	0	0	0	0	0	0	0	19.4	21.0	14.8	12.2	73.1	74.2	72.0	70.9
	1	0	0	0.05	0.05	0	0	0.05	0.05	0	0	0.05	0.05	0	0	0.05	0.05	64.8	65.8	63.3	62.0
timed, $\ell = 2\,\mathrm{ms}$	0.1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	29.8	29.0	26.4	25.7
	0.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	23.1	23.1	20.8	19.8
	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	14.8	14.6	12.4	11.5
timed, $\ell = 5\,\mathrm{ms}$	0.1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	0.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
asap	0.1	0	0	0	0	0	0	0	0	48.0	56.8	32.6	26.0	84.4	88.4	80.4	77.1	94.8	95.2	93.1	91.8
	0.5	0	0	0	0	0	0	0	0	0	0	0	0	44.4	45.6	38.6	35.4	81.5	83.4	80.9	79.8
	1	0	0	0.05	0.05	0	0	0.05	0.05	0	0	0.05	0.05	0	0	0.05	0.05	64.8	65.8	63.3	62.0
tick-based	0.1	92.2	93.6	90.0	88.4	94.7	96.0	92.3	90.7	98.0	98.7	96.6	95.2	99.4	98.0†	97.8†	97.4†	99.8	84.4†	84.4†	84.3†
	0.5	72.2	73.6	70.3	68.5	74.7	75.9	72.6	70.9	79.7	80.6	77.2	75.4	92.3	93.2†	91.0†	89.8†	97.4	89.8†	89.8†	89.8†
	1	47.2	48.6	45.1	43.4	49.7	51.0	47.4	45.8	54.7	56.0	52.5	50.8	72.2	73.0	69.5	67.8	90.7	90.0†	89.0†	88.4†

a pending one before its tick; the next packet then needs no switch, and the PER can fall below the closed form. With the line of sight present, a fraction of the exposed packets fail: 54–99 % at 15 dB for both alphabets, and no cell loses more than 3.1 % at 25 dB. The timing rules are thus consequences of the semantics, and what the IQ level measures is this conditional loss: where a switch inside the preamble starts to cost, how the equalizer fails after it, and how much a line of sight protects.

The served runs of Section VI-C give the latency distribution that replaces the modelled Λ when a controller runs as a separate process: an asap command became visible after a median of 2.2–2.6 ms and a p99 of 5.1–6.0 ms, with rare outliers of up to 17 ms, longer than any latency of the regime $J \leq 1$ ms of Fig. 4. A timed command takes effect at its exact sample when its lead covers the delay to the server, the error of the client's estimate of virtual time and about one page plus its computation (Appendix E): pooled over the repetitions, with a 2 ms lead (E-B used 2 ms), 3.3–5.7 % of the served commands were late; with 4 ms, at most 0.17 %; and with 6–12 ms, from about the asap p99 upward, at most 0.06 %. A timed controller on this path therefore needs a lead of 4 ms or more to stay in the on-time regime of Fig. 4; the timing grid of E-B2 was not rerun through the served path. The clock of the tick-based semantics starts at $t = 0$ and keeps one pending command per panel. With a timer period of the order of a packet, alignment with the frame decides its losses, as the closed form shows; a timer period of seconds, as in the cited 5G NR experiments, would expose at most one packet per period.

In the lab-like geometry (Fig. 4(c), (d)), with the line of sight blocked, the switch changes the channel of RX1 by -15.6 dB and 59°, and 345 payload bits fail for the same switch time. Timed commands with the 2 ms lead lose no packet up to $J = 1$ ms and 22 % at $J = 3$ ms. Asap commands lose 44 % at $J = 1$ ms. Tick-based semantics lose 73 % without jitter and up to 92 % with it. With the line of sight present, no semantics loses a packet at 25 dB; this geometry was not run at 15 dB.

B. E-A: protocol record and time scale

The access point sits 0.5 m off the user line, which keeps the line-of-sight power finite everywhere on the line, and the partition loss gives the RIS room to help. The partition acts on the line of sight alone, although it crosses the path from the access point to every cell in every partitioned geometry. A partition that also attenuated that hop would scale both paths alike and leave the values of the corridor without a partition at 1 GHz: an optimum gain of 0.68 dB and $\tau_{1/2} = 0.106$ s ($0.35\,\lambda/v$). The hypotheses and their ranges were in the emulator's design document before the sweep: GitHub's push log (Section V-D) records it in the initial push of the emulator repository. The plan that adds the decision rules and the evaluation rule (SHA-256 5b59f8f2… with LF line ends) was written before the sweep by the author's record, but committed to the paper repository after it (commit 71fa3f3), so no third party dates those parts. One amendment, made before the full sweep, moved the evaluation of each time step from its start to its midpoint after a 20 s IQ run (exact optimum, $\tau_{\mathrm{SUI}} = 0.1$ s) had shown the start-of-step rule 0.095 dB high and the midpoint 0.002 dB low; the plan discloses it and a short pipeline test seen before it. VA-1 therefore compares the IQ runs with a rule chosen after an IQ result had been seen: it checks the consistency of the evaluator, and it is not a blind test.

The engine's `run_batch` evaluates one step per interpreter iteration, too slow for this sweep, so the points come from a vectorized evaluator in the experiment code that evaluates the same narrowband model. It tabulates the channel of the scene model along the user line on a grid of 1 mm (0.5 mm at 2.4 GHz) and interpolates it complex-linearly, with an error of at most 0.008 dB against the scene model with all cells on, at random times in every geometry; it tabulates the decisions of the position-driven controllers on their 1 cm grid, equal

TABLE XII
E-A OVER EVERY COMPUTED GEOMETRY: HALF-RETENTION DELAY $\tau_{1/2}$ OF THE EXACT OPTIMUM (95 % INTERVAL OVER 20 BATCHES), ALSO IN UNITS OF λ/v ($v = 1$ m/s), RECEIVED POWER WITH THE INSTANTANEOUS OPTIMUM $P(0)$, AND ITS GAIN OVER RIS ABSENT (MEAN OF PER-BATCH DB GAINS). h: DISTANCE OF THE CELL ROW FROM THE USER LINE; PART.: PARTITION LOSS, APPLIED TO THE LINE OF SIGHT, NOT THE RIS HOPS. *: PRIMARY GEOMETRIES OF SECTION VII-B. SIM-SREMU.

f_c (GHz)	h (m)	Part. (dB)	$\tau_{1/2}$ (s)	95 % interval (s)	$\tau_{1/2}v/\lambda$	$P(0)$ / gain (dBm / dB)
1.0	1	0	0.128	0.126–0.129	0.43	−40.6 / 1.01
1.0	2	0	0.106	0.105–0.107	0.35	−40.9 / 0.68
1.0	2	10	0.112	0.111–0.113	0.37	−49.5 / 2.06
1.0*	2	20	0.126	0.125–0.127	0.42	−56.0 / 5.61
1.0	2	30	0.163	0.162–0.165	0.55	−59.0 / 12.54
1.0	4	0	0.088	0.087–0.089	0.29	−41.2 / 0.37
1.0	4	20	0.107	0.107–0.108	0.36	−58.2 / 3.41
1.0	6	0	0.082	0.081–0.082	0.27	−41.3 / 0.23
2.4	2	10	0.050	0.049–0.050	0.40	−57.2 / 2.03
2.4*	2	20	0.054	0.054–0.055	0.43	−63.7 / 5.48
2.4	2	30	0.062	0.062–0.063	0.50	−67.1 / 12.05
2.4	4	20	0.046	0.046–0.047	0.37	−65.8 / 3.36

to the live controllers' decisions at 50 400 sampled barriers; it runs URAP and the bandit barrier by barrier with the engine's random keys; and it evaluates each 5 ms step, split at switch times, at its midpoint. With the start-of-step rule instead, it matches `run_batch` within 0.0003 dB in the VA-1 configurations. VA-1 does not test the heuristic, the bandit, the 2.4 GHz carrier, the other geometries or the time series of the power; at the received power of its runs, noise moves the mean power by less than 10^{-6} dB, so its noise-on rows repeat the noise-off test. Every point follows one random-waypoint trajectory (one seed), shared by all controllers, at one speed, and the RIS is a single row of $\lambda/4$ cells in a planar geometry; other trajectories, speeds or two-dimensional panels may change the constants.

The heuristic of [13], as run, takes the OFF states as base, $b = h_{\mathrm{d}} + \sum_n Z_{\mathrm{off},n}$, and $d_n = Z_{\mathrm{on},n} - Z_{\mathrm{off},n}$ as the change when cell n is switched on. A cell is on when its relative phase $\delta_n = \arg b - \arg d_n$ lies in an arc of width Θ at offset ψ from one edge, $(\delta_n)_{2\pi} < \Theta - \psi$ or $(-\delta_n)_{2\pi} < \psi$, with $(\cdot)_{2\pi}$ reduced to $[0, 2\pi)$. The controller evaluates $|b + \sum_{n\in A} d_n|^2$ on its channel model for $K = 24$ candidates, $\Theta_l = l\pi/(2L)$ for $l = 1, \ldots, L$ and $\psi = j\Theta/(m+1)$ for $1 \le j \le m \le M$ with $L = 4$ and $M = 3$, and keeps the best. The letter fixes K but not the values of ψ; this grid is our choice.

Table XII lists every computed geometry: $\tau_{1/2}$ spans 0.082–0.163 s at 1 GHz and 0.27–0.55 λ/v over all geometries; at a fixed carrier and offset it rises with the partition loss, and at a fixed loss it falls with the offset h. The constant thus belongs to the geometry and to the balance between the line of sight and the RIS path. For isotropic scattering the field and envelope correlations fall to one half after 0.24 and 0.18 λ/v [72]. The rule-of-thumb coherence time $\sqrt{9/(16\pi)}/f_D = 0.42\,\lambda/v$ lies close to the E-A constant, but it is the geometric mean of $1/f_D$ and the heuristic half-correlation time $9/(16\pi f_D)$, not a correlation level. The ratio of the two half-lives, 0.429 (0.428–0.430), lies 3 % above the wavelength ratio 0.417; the corridor is the same in metres at both carriers, so it is not the same geometry in wavelengths and exact scaling is not expected. The ranges fixed before the sweep were 0.1–1 s for H-A1 and 0.30–0.55 for H-A2, so their support is a weak test. What E-A adds to the classical scaling is the constants for these geometries, the retention metric and the result for controllers without positions: besides the bandit (H-A3 supported), URAP gains 0.12 dB, and the heuristic of [13] reaches 4.45 dB at the shortest delay.

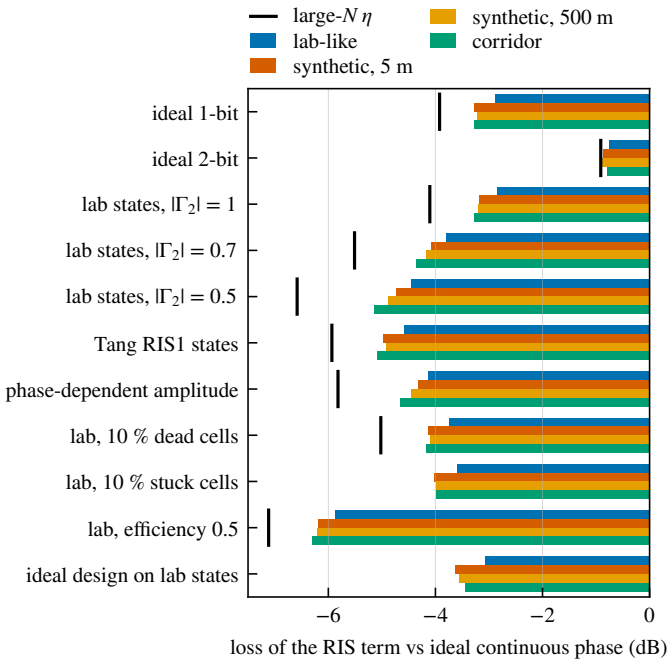


Fig. 11. Loss of the optimized RIS term relative to ideal continuous phase for each alphabet or defect in the lab-like geometry, the synthetic panel with a near and a far receiver and the blocked corridor; black ticks: large-N closed form. Data classes: SIM-sremu (optimized runs), ANALYTIC (ideal alphabets, closed form), ASSUMED (laboratory state amplitudes), REPORTED (published states of Tang et al.). $|\Gamma_2|$: magnitude of the second laboratory state, the other having unit magnitude.

C. E-C per geometry

Fig. 11 and Table XIII give the loss of the optimized RIS term relative to ideal continuous phase in four settings, per alphabet or defect, with the large-N closed-form loss. The prior work they agree with concerns discrete and phase-dependent states [8], [11]; the laboratory state amplitudes are ASSUMED as in Section VII-A.

D. FM-band dimensioning (E-D)

This subsection is model-based dimensioning, not validated. Can a focused RIS restore an FM broadcast link of 52 km at 100 MHz behind an obstruction? The link and the panel of this scenario come from the unpublished ANTS 2025 submission (Section II); this dimensioning replaces its result. With a flat-earth two-ray direct path, a coherent knife edge and ground on both RIS hops, the aperture needed for a RIS path as strong as the direct path is 117–6522 m^2 over the receiver heights and distances of Fig. 12. At grazing obstruction, over the same heights and distances, the RIS path of a 56.25 m^2 panel is 17.5–41.3 dB weaker than the direct path. Over all map entries with this direct path, for two placements of the

TABLE XIII
NON-IDEALITY ABLATION (E-C): LOSS OF THE OPTIMIZED RIS TERM RELATIVE TO IDEAL CONTINUOUS PHASE, IN dB, PER GEOMETRY, AND THE LARGE-N CLOSED-FORM LOSS η. SIM-SREMU; IDEAL ALPHABETS ANALYTIC; LAB AMPLITUDES ASSUMED; TANG STATES REPORTED. $|\Gamma_2|$: MAGNITUDE OF THE SECOND LABORATORY STATE, THE OTHER HAVING UNIT MAGNITUDE; —: NO CLOSED FORM FOR STUCK CELLS, WHOSE FIXED TERM DEPENDS ON THE RANDOM DRAW.

Alphabet or defect	Lab-like	Synth. 5 m	Synth. 500 m	Corr.	η
ideal 1-bit	−2.89	−3.27	−3.22	−3.28	−3.92
ideal 2-bit	−0.75	−0.87	−0.89	−0.79	−0.91
lab states, $\|\Gamma_2\| = 1$	−2.86	−3.18	−3.20	−3.29	−4.10
lab, $\|\Gamma_2\| = 0.7$	−3.79	−4.09	−4.17	−4.36	−5.51
lab, $\|\Gamma_2\| = 0.5$	−4.46	−4.74	−4.89	−5.15	−6.58
Tang RIS1	−4.59	−4.97	−4.91	−5.09	−5.93
phase-dep. amplitude	−4.14	−4.32	−4.45	−4.66	−5.82
lab, 10 % dead	−3.75	−4.13	−4.09	−4.17	−5.02
lab, 10 % stuck	−3.60	−4.02	−3.98	−3.98	—
lab, efficiency 0.5	−5.87	−6.19	−6.21	−6.30	−7.12
ideal design, lab states	−3.07	−3.63	−3.57	−3.45	—

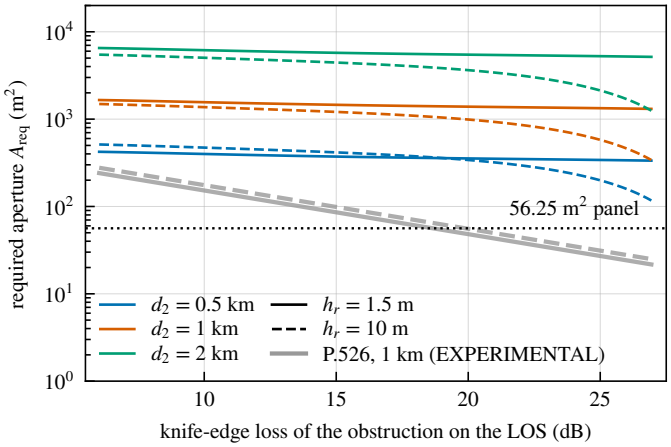


Fig. 12. Model-based dimensioning, not validated: aperture that a focused RIS needs for a path as strong as the direct path, versus the knife-edge loss of the obstruction, for three RIS-to-receiver distances and two receiver heights; the dotted line marks the panel size; grey: the experimental P.526 direct path. Data classes: SIM-sremu, ANALYTIC (panel aperture), ASSUMED (ground constants).

panel and with or without ground on its hops, the panel suffices in 0 of 1224. A four-ray direct path over flat ground, a model-form sensitivity, gives 23 entries. The picture changes with the direct-path model: with the experimental ITU-R P.526 smooth-earth path of `sremu`, which is weaker, the same panel suffices in 844 of 1224 entries. That module is not validated, so the dimensioning question is decided by the direct-path model rather than by the RIS.

E. E-E: GNU Radio runs

Besides E-E, two studies ran with external processes: the frequency-offset sweep between two processes over the native data plane (V-CFO-3) and the consistency runs of E-A with an external controller and receiver (VA-1). The GNU Radio DQPSK modem uses 4 samples per symbol, and the partition attenuates the direct path of the E-E scene. Table XIV lists the static runs of Section VII-D, one fresh run with a fresh modem per RIS configuration; the block error rate counts uncoded 200-bit blocks. Where at least one hundred symbol errors were counted with the link up, the symbol error rate (SER) implies an implementation loss of 0.55–0.56 dB against coherent DQPSK theory at the estimated E_s/N_0; with the link lost, the implied loss is larger.

TABLE XIV
GNU RADIO DQPSK MODEM OF STOCK BLOCKS THROUGH THE RAW REQUEST–REPLY MODE (E-E), ONE FRESH RUN PER RIS CONFIGURATION: FOCUSING, FOCUSING WITH A SEEDED FRACTION q OF ITS CELLS INVERTED, UNIFORM (EVERY CELL IN STATE 0) AND RANDOM. AF: ISOTROPIC ARRAY-FACTOR PREDICTION RELATIVE TO FOCUSING; GAIN: LEAST-SQUARES GAIN OF THE IQ THAT GNU RADIO RECEIVED, RELATIVE TO FOCUSING (AF AGAINST GAIN IS AN INTERNAL CONSISTENCY CHECK, NOT A MEASUREMENT); SNR: ESTIMATED FROM THE RECEIVED IQ OVER THE SAMPLE BANDWIDTH; SER: GNU RADIO'S SYMBOL DECISIONS, 69 960 SCORED PER CONFIGURATION; BLER: FRACTION OF THE 699 UNCODED 200-BIT BLOCKS (100 CONSECUTIVE DECISIONS, NO CODING, NO CRC) WITH AN ERROR. DATA CLASSES: SIM-SREMU (EMULATED IQ AND THE DECISIONS ON IT), ANALYTIC (AF).

RIS configuration	AF (dB)	Gain (dB)	SNR (dB)	SER	BLER
Focusing	0.00	0.00	15.3	0	0
$q = 0.10$	−1.98	−1.94	13.4	0	0
$q = 0.20$	−4.43	−4.34	11.0	0	0
$q = 0.25$	−6.20	−6.05	9.3	0	0
$q = 0.30$	−8.01	−7.79	7.6	2.9×10^{-5}	1.4×10^{-3}
Uniform	−8.32	−8.06	7.3	5.7×10^{-5}	2.9×10^{-3}
$q = 0.35$	−10.95	−10.57	4.8	2.3×10^{-3}	0.11
$q = 0.40$	−13.83	−13.25	2.1	3.3×10^{-2}	0.78
$q = 0.50$	−19.02	−18.39	−3.0	0.45	1.00
Random	−23.71	−22.73	−7.4	0.59	1.00

The change point of the switching run is the maximum-likelihood change point of the received IQ, and in every run every engine and transport counter stayed zero.

In three sweeps of single runs on a laptop (AMD Ryzen 9 8945H, WSL2, host load not controlled), the lockstep loop with GNU Radio advanced at an RTF of 0.60–1.28 at 1 MS/s and 0.39–0.44 at 5.76 MS/s; on an AWS c7i.48xlarge instance it reached 2.88 at 1 MS/s.

srsRAN and OpenAirInterface were analysed but not run. srsRAN 4G pads its transmit stream to the end of the receive block it is about to read and no further, while lockstep needs the transmit stream one fractional-delay look-ahead beyond the received samples; a lockstep srsRAN pair therefore needs a per-peer transmit delay and partial receive replies, and srsRAN 4G has no package for the Ubuntu release used here, so it needs a source build; an attach test at LTE sample rates would run slower than the wall clock. The OpenAirInterface RF simulator would need a new transport for its own TCP framing [31]. Both are future work.

APPENDIX G
ORACLE DETAILS

The tests of `sremu` form four tiers. Analytic oracles (identifiers PH, RT and IO) compare the code with closed forms or exact properties in double or exact arithmetic and run in continuous integration. Statistical oracles (ST) run seeded Monte Carlo experiments with stated confidence levels. Code-to-code references (G1–G8) compare the emulator with stored outputs of MATLAB R2026a scripts (those of the RIS kernel were written for K1); MATLAB is not needed to run them,

and their agreement shows agreement between two codes, not validity. Validation and reproduction studies (V) use published or measured data.

The closed forms live in a separate test package that was written from independent derivations and scripts, not from the emulator's code, and that may not import the physics, channel or impairment packages; a static check of the imports and a check in a fresh interpreter enforce this. The physics oracles cover the Friis equation, image theory for plates and disks, the monostatic radar cross section of a plate, the far-field gain of a focused panel, focusing saturation, Fresnel reflection and the two-ray model, the knife-edge Fresnel integral, the losses of finite alphabets and of the phase-dependent amplitude model, and exhaustive searches for the exact optimum. The runtime oracles cover exact sample counts and round trips at several incommensurate rates, the switch sample of a distant receiver, bitwise invariance under chunking, cuts, skipped pages and thread counts, closed-form transition profiles and the paced-mode counters; property tests draw random rational rates, times and periods. The interface oracles feed malformed frames and messages, check the size and element limits and the loopback rule, and check the structure of the SigMF recordings. The statistical oracles test the stationary distribution of the random-waypoint model, the autocorrelation and level-crossing rate of the diffuse term against Clarke's model [72], and the Monte Carlo losses of quantization and failures.

Several guards have a negative-control twin: a test that runs a deliberately wrong variant and must reject it, for example a kernel without the factor j, a fractional-delay filter that is too short, or a replay that drops a command. A guard that accepts its wrong variant is broken. Section V and Appendix D summarize every check with its status as recorded in the test record of commit 1e1fcab (Appendix H); a spec identifier counts as passed when at least one of its implementation tests passed and none failed or was skipped, and the tests of the rendering path count once per backend.

Tables XV and XVI list every verification identifier cited in this paper with the property it checks, its oracle or reference and its tolerance. Apart from VA-1 and VA-4 on the batch path, Sections III and IV cite no identifiers in place, so the tables also name, for each identifier, the subsections of those sections whose statements it checks, taken from the evidence that the pipeline's claim file records for each statement and from the identifiers cited there. They are generated from a transcription that the pipeline checks against the emulator's specification, its table of oracle values and the docstrings of the tests that carry each identifier: a number found in none of them, an identifier cited without a row and an identifier that did not pass in the test record all stop the pipeline. The VA identifiers (VA-1, VA-3, VA-4 and VA-5, the complete list of the specification) are consistency checks between the engine's paths and run in the test record beside the four tiers.

APPENDIX H
REPRODUCIBILITY

The paper is built from two repositories, the emulator and the paper, by a Python pipeline rather than by hand. The paper's lock file names the emulator commit that holds the results. In the emulator repository, `python tools/tasks.py paper-junit FILE` runs the test tiers and writes a test record with the commit and the state of the working tree. In the paper repository, the single command `python tools/build_paper.py --mode final --profile full --junit FILE` then rebuilds every stale figure and table and compiles the paper. It refuses to produce a PDF unless the installed emulator is the locked commit, both working trees are clean, every identifier of both verification gates passed in a test record of the locked code, every figure is current, every number is a macro written by the pipeline and every claim is supported. A figure is current when its stored content key, a hash of the emulator commit, the sweep specification, the input files and the figure script, matches. The build does not rerun studies: a result file enters the key as recorded, so a result produced at another commit, or from a tree whose state was not recorded, is accepted, and Table XVII shows its commit and tree state. The constants and defaults quoted in Sections III and IV and Appendices B and C are read from the locked code at build time, so the text cannot drift from the code. Every run directory carries the manifest of Section III-A, and a replay reproduces its received samples bit for bit. The build checks the lock commit, the test record and the figure content keys; the results themselves record the commit, tree state and host they were produced on, and Table XVII lists them. Both repositories are public (section Code and Data Availability).

The results produced at emulator commit bf08268 from a clean tree are V-TANG, V-CFO, V-FM (E-D), E-A, E-B (lab-like), E-C, BER, VA-1, E-B (arc geometry), E-F latency, B2-F45, B3, B4, B4-classic and E-E (GNU Radio); all of them except the benchmarks come from one regeneration job on an AWS c7i.48xlarge instance, and the benchmarks are measurements on dedicated hosts, run as separate jobs (Appendix E). The test record was made at 1e1fcab, and the build checks that the installed emulator's package source (a hash over all its modules) equals the one these results record. Each result keeps the commit at which it was produced (Table XVII); for the other results these are the test record (1e1fcab, clean); V-LAB (7a63287, clean); V-LAB power analysis (9dca79f, uncommitted changes recorded); F3(a) values, F6 values and cell-model sensitivity (1e1fcab, tree state not recorded); B1 and B2 (309c633, dirty flag recorded, file tree equal to the commit's); B6 (06cdf7a, clean); E-E throughput (commit not recorded). V-LAB is not rerun, because its analysis plan is locked: it ran once, at 7a63287 from a clean tree, on predictions computed before the lock. A check at 1e1fcab found the V-LAB protocol and analysis files byte-identical to those of 7a63287 and the protocol hash equal to the lock; emulator modules that the analysis imports have changed since, so we do not claim that a rerun would reproduce it.

APPENDIX I
DATA PROVENANCE

Every input and every figure carries data classes: MEAS (measured data), DIG (digitized from a figure, with its digitization uncertainty), REPORTED (quoted from a publication),

ASSUMED (chosen where no data exist, such as the amplitudes of the laboratory states), ENG (an engineering choice made for a study, such as receiver positions or the transition shape), SIM-sremu (outputs of this emulator), ANALYTIC (closed forms) and MATLAB (code-to-code references). Run manifests list the hash and data class of every input, and every figure caption names the classes it shows. Input files are copied into the repositories with their hashes and provenance, with one exception: the raw input file of the laboratory map of Section V-D is read from a location outside the repositories that an environment variable names, and the tests that need it are skipped when it is absent; its per-cell values are in the published V-LAB result files. Response tables keep their own provenance (Section III-D). The laboratory measurement of Section V-D is a measurement collected at IIT Bhilai and enters as a digitized map (DIG); Appendix D states its origin. Process records carry no physical quantity: GitHub's push log of the emulator repository, quoted in Section V-D, is kept as an exported copy with its hash. Table XVII lists the producing commit, tree state, host and date of every result. The digitized measurements of Tang et al. enter as DIG with the reported link budget (REPORTED); Appendix D-C breaks their comparison with the emulator (Section V-C) down by geometry.

TABLE XV

PHYSICS IDENTIFIERS (PH) AND MATLAB CODE-TO-CODE REFERENCES (G): THE PROPERTY CHECKED, ITS ORACLE OR REFERENCE, AND THE ACCEPTANCE TOLERANCE, TRANSCRIBED FROM THE EMULATOR'S SPECIFICATION AND TEST MODULES (THE PIPELINE REQUIRES EACH NUMBER TO APPEAR IN THEM). TOLERANCES IN DB, DEGREES, HZ OR SAMPLES ARE ABSOLUTE; REL.: RELATIVE TO THE REFERENCE MAGNITUDE; BITWISE: IDENTICAL SAMPLES. SEC.: SUBSECTIONS OF SECTIONS III AND IV WHOSE STATEMENTS THE IDENTIFIER CHECKS (THOSE SECTIONS CITE FEW IDENTIFIERS IN PLACE); –: NONE. TESTS: IMPLEMENTATION TESTS IN THE TEST RECORD OF COMMIT 1E1FCAB THAT CARRY THE IDENTIFIER. EXP.: EXPERIMENTAL TIER, OUTSIDE THE RECORD.

Id	Property	Oracle or reference	Tolerance	Sec.	Tests
PH-01	Free-space line of sight (Friis)	closed form: -63.329 dB at 3.5 GHz, 10 m; -106.768 dB at 100 MHz, 52 km	rel. 10^{-12}	IV-A	4
PH-02	Kernel K1 equals published forms (point cells)	Tang et al. Eq. 12, ETSI far field with area gain, Ellingson	rel. 10^{-12}	IV-B	13
PH-03	N^2 law of a focused far-field panel	Tang et al. Eq. 8: $N = 100$, 400, 1600 at $\lambda/2$, 2 km	$\pm$0.01 dB; slope 12.04 $\pm$0.01 dB per $4\times$	–	2
PH-04	Power invariant under aperture refinement	1 m^2, 3.5 GHz: -115.280 dB far, any N; -88.128 dB near, converged patches	$\pm$0.01 dB far; $\pm$0.10 dB near	IV-B, IV-C	11
PH-05	Image theory and the Huygens factor j	40° plate of 12 m vs image; disk on axis vs closed form	0.3 dB, 3°; 0.01 dB	IV-B	3
PH-06	Monostatic radar cross section of a plate	1 m^2 at 400 m: -126.067 dB	$\pm$0.01 dB	–	1
PH-07	Near-field focusing saturation, passivity	$d = 1$ m: -23.864 dB ($L = 1$ m), -6.021 dB ($L \to \infty$); $\lvert h\rvert \le 0.5$	$\pm$0.01 dB; bound exact	–	5
PH-08	Reciprocity of the full scene	LOS, ground, two walls, knife edge, RIS with a symmetric table	rel. 10^{-12}	IV-A	20
PH-09	Visibility of the panel	node behind the panel	RIS term exactly 0	IV-B	5
PH-10	Flat ground, Fresnel coefficients	two-ray loss 139.10 dB (52 km, 100 MHz); Brewster angle 26.565°; PEC	$\pm$0.01 dB; 10^{-12}; exact	IV-A	17
PH-11	Single knife edge $F(\nu)$	$\lvert F(0)\rvert = 0.5$; approximation $J(\nu)$ vs exact loss; $\nu = 10$: 32.95 dB	10^{-12}; 0.15 dB; $\pm$0.01 dB	IV-A	29
PH-12	Smooth-earth P.526 path (experimental tier)	regression: excess loss 37.14 and 20.28 dB at 52 km	$\pm$0.01 dB	–	exp.
PH-13	Large-N quantization loss	$\eta(b) = -3.922$, -0.912, -0.224 dB; lab pair -4.105 dB; Monte Carlo	$\pm$0.001 dB; Monte Carlo $\pm$0.15 dB	IV-D	11
PH-14	Phase-dependent amplitude model	$(\beta_{\min}, \alpha) = (0.2, 1.6)$: -5.508 dB; $(0.8, 1.6)$: -1.085 dB	$\pm$0.01 dB	–	5
PH-15	Efficiency and dead cells	$\eta = 0.5$: -3.0103 dB; 10% dead cells, $N = 1024$: -0.915 dB	10^{-9}; $\pm$0.10 dB	IV-D	6
PH-16	Phase-wrap invariance	phases shifted by whole turns	$\lvert\Delta h\rvert \le 10^{-15}\,\lvert h\rvert$	–	14
PH-17	Geometric Doppler shift	LOS at 30 m/s, 3.5 GHz: 350.242 Hz; RIS path: 325.192 Hz	LOS: 10^{-6} rad, $\pm$0.001 Hz; RIS path: $\pm$0.01 Hz	IV-E	10
PH-18	Snapshot interpolation and render tiles	brute force per element and sample, three scenes	$\lvert\Delta h\rvert/\lvert h\rvert \le 10^{-3}$	IV-E	27
PH-19	Invariance to snapshot period and chunking	T_u = auto, auto/2, auto/4; client chunks of 0.25, 1, 3.7 ms	rel. 2×10^{-3}; chunks bitwise	IV-E	9
PH-20	Fractional delay, wideband rendering	every filter row in band; known delay of 37.4 samples	≤ -60 dB; 0.01 samples	IV-E	41
PH-21	Thermal noise power, independence	-113.975 dBm at 1 MS/s, $F = 0$ dB; correlation across receivers	$\pm$0.05 dB; $\lvert\rho\rvert \le 0.016$	IV-E	6
PH-22	Linearity, time shift, seeds	exact algebraic identities	10^{-12}	–	10
PH-23	Energy bound of a panel	scattered vs incident power, every tested configuration	scattered $\le$ incident	–	21
PH-24	Exact binary optimum	brute force: $N = 20$ (2^{20} configurations), 100 positions	rel. 10^{-12}	IV-D	2
PH-25	Binary decomposition $\Gamma = m \pm d$	decomposed vs direct; lab pair, unit magnitudes assumed: $\lvert d\rvert$: -0.18 dB, $\lvert m\rvert$: -13.86 dB	10^{-12}; $\pm$0.01 dB	IV-D	6
PH-26	Point vs patch cell model	16×16 panel, 3.5 GHz: specular; 45° retro, 8.11 dB in the far field	0.05 dB; $\pm$0.1 dB	IV-C	7
PH-27	Occluders and partitions	segment-polygon cases; partition of L dB on the LOS	exact	IV-A	24
PH-28	Extension of incidence-angle tables	symmetry and specular entry of the extended table	exact	III-D	2
PH-29	General finite-alphabet optimum	brute force: $S = 4$, $N = 10$, 50 instances	rel. 10^{-12}	IV-D	1
PH-30	Quantizer	per-element brute force, unequal amplitudes	exact	IV-D	3
PH-31	Response-table validators	passivity, reciprocity, grid density, domain, provenance	exact or raises	III-D	36
PH-32	Ground on the RIS hops	$\Gamma_g = 0$ gives the free-space term; four hop variants	10^{-12}	IV-A	9
G1	Per-element K1 sum	toolbox primitives; lab-like layouts, geometries of Tang et al.	max. rel. 10^{-9}	–	17
G2	Steered beam pattern	phased.RISReflector, 16×16 at $\lambda/2$, constant removed	$\pm$0.5°; 0.05 dB	–	3
G3	One-bit beam pattern	mirror lobe at 41.15°, -0.97 dB	$\pm$0.25°; $\pm$0.10 dB	–	3
G4	Frequency offset	comm.PhaseFrequencyOffset, arbitrary chunk sizes	10^{-9} of the RMS	IV-E	3
G5	Thermal noise	comm.ThermalNoise, 290 K, 1 MS/s	$\pm$0.01 dB	IV-E	3
G6	Fading statistics	comm.RayleighChannel autocorrelation and level crossings	as ST-02	–	1
G7	ETSI far-field closed form	focused panel at 100 MHz; physical-optics plate bound	1 dB; bound + 0.01 dB	–	2
G8	Two-ray channel	phased.TwoRayChannel at 52 km	$\pm$0.05 dB	–	1

TABLE XVI

RUNTIME (RT), INTERFACE (IO), STATISTICAL (ST), ENGINE-PATH CONSISTENCY (VA) AND FREQUENCY-OFFSET (V-CFO) IDENTIFIERS, AS IN TABLE XV. TOLERANCES IN DB, DEGREES, HZ OR SAMPLES ARE ABSOLUTE; REL.: RELATIVE TO THE REFERENCE MAGNITUDE; BITWISE: IDENTICAL SAMPLES. SEC.: SUBSECTIONS OF SECTIONS III AND IV WHOSE STATEMENTS THE IDENTIFIER CHECKS (THOSE SECTIONS CITE FEW IDENTIFIERS IN PLACE); –: NONE. TESTS: IMPLEMENTATION TESTS IN THE TEST RECORD OF COMMIT 1E1FCAB THAT CARRY THE IDENTIFIER. STUDY: ACCEPTED IN THE STUDY RECORD (APPENDIX I), NOT IN THE TEST RECORD.

Id	Property	Oracle or reference	Tolerance	Sec.	Tests
RT-01	Exact sample counts	5880 S/s, 23.04 and 30.72 MS/s together; awkward durations	exact	III-B	13
RT-02	Time round trips	k up to 2^{40} at three rates; property tests	exact	III-B	14
RT-03	Invariance to chunks, epochs, barriers, threads	client chunks of 1, 7, 1000 samples; 1, 4, 16 threads	bitwise	III-B	24
RT-04	Page-size invariance	page targets of 0.25, 0.5, 1 ms	rel. 10^{-9}	III-B	8
RT-05	Switch sample of each receiver	$\mathrm{to_ticks}(t_{\mathrm{sw}} + \tau)$ for a receiver at 2 km, 23.04 MS/s	exact	III-B	5
RT-06	Late commands	applied on arrival with logged lateness, or rejected	exact	III-B	6
RT-07	Same-time ties	later sequence number wins, both logged	exact	III-B	6
RT-08	Barrier determinism	controllers sleeping 0 vs 50 ms; random rates and periods	bitwise; lateness 0	III-A, III-B	20
RT-09	Oscillator-offset continuity	500 Hz at 5880 S/s across block boundaries	$\pm 10^{-9}$ rad	IV-E	14
RT-10	Skip invariance	receiver inactive for 0.3 s, then active	bitwise	III-B	8
RT-11	Known delay through the engine	37.4 samples	0.01 samples	IV-E	6
RT-12	Reconfiguration transitions	linear 20 μs ramp; re-command during a transition	10^{-9}; 10^{-3}	III-B	11
RT-13	Transmit commitment and stalls	end of burst, idle-until, stall report, zero fill	as specified	III-B	9
RT-14	Keyed random streams	an added node leaves other variates unchanged; two backends	bitwise	IV-E	21
RT-15	Independence of the wall clock	static import check; patched clock	identical output	III-B	17
RT-16	Replay	recorded TX and commands reproduce the RX samples	bitwise	III-A, III-B	18
RT-17	Lookahead violation	action earlier than $t_b + L_{\min}$	error raised	III-B	8
RT-18	Paced-mode counters	injected sleep load, late TX burst, unread RX stream	exact counts	III-B	10
RT-19	Backend equivalence through the engine	lab walk, vehicle and FM scenes	rel. 10^{-9}; RMS 10^{-6}	–	4
IO-01	Malformed frames and messages	10^4 malformed frames; malformed JSON	no crash; exact counts	III-C	22
IO-02	Two OS processes over native ZMQ	TX and RX processes vs in-process lockstep	bitwise	III-A	3
IO-03	Socket command at the exact sample	commands over the control socket in lockstep	lateness 0	III-A, III-B, III-C	12
IO-04	Size, element and bind limits	1 MiB requests, 65,536 elements, loopback binds	refused and counted	III-C	27
IO-05	SigMF recordings	required keys, annotation order, collection hashes	pass	III-A, III-B	30
IO-06	API status codes	timeout, overflow, late command, time and sequence errors	exact	III-A, III-C	14
IO-07	Compatibility wrappers	in-house BPSK scripts through the wrapper calls	2389 samples, exact	–	11
IO-08	Raw request-reply mode	stub peers, both request formats	sample-exact in lockstep	III-C	11
ST-01	Random-waypoint law	$l = 10$ m, $v = 1$ m/s, 10^4 periods vs stationary law	KS $D \leq 0.02$; mean leg $\pm 2.5\%$	IV-A	4
ST-02	Diffuse fading statistics	$f_D = 50$ Hz: autocorrelation vs J_0; level crossings	0.05; 10%; KS 0.01	IV-A	7
ST-03	Quantization and failure Monte Carlo	as PH-13 and PH-15, more trials	± 0.15 dB; ± 0.10 dB	IV-D	4
ST-04	BER through the emulator	BPSK, QPSK: 6 points, ≥ 100 errors or 2×10^6 bits	theory inside 99% interval	–	12
ST-05	Frequency estimator vs Cramér–Rao bound	periodogram at 10 and 20 dB SNR	RMSE ≤ 1.5 bound	–	3
VA-1	Served IQ path vs batch evaluation (E-A)	external controller and RX application, 1000 s runs	0.02 dB, noise off; CI, noise on	III-B	1
VA-3	Exhaustive optimum bounds every controller	$N = 20$ (2^{20} configurations), 29 positions	rel. 10^{-12}	–	4
VA-4	IQ path equals batch path	static lab-like scene, noise off, no signaling delay	rel. 10^{-9}	III-B	2
VA-5	Sample-exact intra-frame switch	earliest affected sample; earlier samples vs no switch	exact; bitwise	–	13
V-CFO-1	Frequency offset, noise off	BPSK burst of 2389 samples at 5880 S/s, offsets to ± 1400 Hz	10^{-6} Hz; epochs identical	–	5
V-CFO-3	Same sweep, two OS processes over ZMQ	unwrapped-phase estimator; output vs V-CFO-1	slope 1 ± 0.002, $\lvert\text{intercept}\rvert < 2$ Hz; bitwise	–	1
V-CFO-4	Frequency estimator with physical noise	SNR 0, 10, 20 dB; 200 trials at 7 offsets	RMSE ≤ 1.5 bound at 10, 20 dB	–	study

TABLE XVII

PROVENANCE OF THE RESULTS QUOTED IN THIS PAPER, AS THE RESULT FILES RECORD IT: EMULATOR COMMIT, STATE OF ITS WORKING TREE, HOST, DATE (UTC) OF THE RUN, AND THE RECORD (SREMU `VALIDATION/*/RESULTS`, PAPER `SWEEPS/RESULTS` OR `FIGURES/DATA`). FOR AN AGGREGATE THE COMMIT IS THE ONE ITS AGGREGATION STEP RECORDED; ITS POINT RUNS ARE LISTED IN ITS OWN PROVENANCE. FLAGGED: THE RUN RECORDED A DIRTY FLAG, BUT ITS UPLOADED FILE TREE EQUALS THE COMMIT'S. N/R: NOT RECORDED. V-TANG, V-CFO AND V-FM (E-D) RECORD THE PLATFORM, NOT THE INSTANCE; THEIR HOST IS THAT OF THE REGENERATION JOB WHOSE COMMIT AND GIT-STATE FILE THEY RECORD. THE V-LAB ANALYSIS IS NOT RERUN BY DESIGN (ITS PLAN IS LOCKED); APPENDIX H EXPLAINS WHICH RESULTS COME FROM THE REGENERATION JOB AND WHICH KEEP THEIR OWN COMMIT.

Result	Used in	Commit	Tree	Host	Date	Record
Test record	Table VII	1e1fcab	clean	laptop (WSL2, Ryzen 9 8945H)	2026-10-12	`sremu_paper.xml`
V-TANG	Fig. 2(b), Table VIII	bf08268	clean	AWS c7i.48xlarge	2026-10-12	`20261012T025951Z_vtang`
V-CFO	Fig. 8(a, b)	bf08268	clean	AWS c7i.48xlarge	2026-10-12	`20261012T025950Z_vcfo`
V-FM (E-D)	Fig. 12	bf08268	clean	AWS c7i.48xlarge	2026-10-12	`20261012T025951Z_vfm`
V-LAB	Table IX, Fig. 7	7a63287	clean	laptop (Windows)	2026-10-11	`20261011T003044Z_vlab_unblinded`
V-LAB power analysis	Section V-D	9dca79f	dirty	n/r	2026-10-10	`20261010T223000Z_vlab_power`
F3(a) values	Fig. 2(a)	1e1fcab	n/r	laptop (Windows)	2026-10-12	`figures/data/F3.meta.json`
F6 values	Fig. 9	1e1fcab	n/r	laptop (Windows)	2026-10-12	`figures/data/F6.meta.json`
Cell-model sensitivity	Section IV-C, Appendix C	1e1fcab	n/r	laptop (Windows)	2026-10-12	`sweeps/results/sensitivity_cellmodel`
E-A (aggregate)	Fig. 5(a, b)	bf08268	clean	AWS c7i.48xlarge	2026-10-12	`sweeps/results/E-A`
E-B, lab-like (aggregate)	Fig. 4(c, d)	bf08268	clean	AWS c7i.48xlarge	2026-10-12	`sweeps/results/E-B1`
E-C (aggregate)	Table XIII	bf08268	clean	AWS c7i.48xlarge	2026-10-12	`sweeps/results/E-C`
BER (aggregate)	Fig. 8(c)	bf08268	clean	AWS c7i.48xlarge	2026-10-12	`sweeps/results/E-F/ber`
VA-1 (rows)	Fig. 5(c)	bf08268	clean	AWS c7i.48xlarge	2026-10-12	`sweeps/results/E-A-IQ`
E-B, arc geometry	Fig. 4(a, b)	bf08268	clean	AWS c7i.48xlarge	2026-10-12	`regen_2026-10-12T02:54:47Z`
E-F latency	Section VI-A	bf08268	clean	AWS c7i.48xlarge	2026-10-12	`sweeps/results/E-F/latency`
B1	Appendix E	309c633	flagged	AWS c7i.48xlarge	2026-10-11	`20261011T083749Z_b1_slot16`
B2	Fig. 10(a–e)	309c633	flagged	AWS c7i.48xlarge	2026-10-11	`20261011T084048Z_b2_server`
B2-F45	Fig. 10(f), Section VI-A	bf08268	clean	AWS c7i.48xlarge	2026-10-12	`20261012T032501Z_b2_server`
B3	Fig. 3(a, b)	bf08268	clean	AWS c7i.8xlarge	2026-10-12	`20261012T025455Z_b3_server`
B4	Fig. 3(c, d), Section VI-C	bf08268	clean	AWS c7i.8xlarge	2026-10-12	`20261012T025715Z_b4_paced`
B4-classic	Appendix E	bf08268	clean	AWS c7i.8xlarge	2026-10-12	`20261012T033849Z_b4_paced_classic`
B6	Appendix E	06cdf7a	clean	laptop (Windows)	2026-10-11	`20261011T185526Z_b6_laptop`
E-E (GNU Radio)	Table XIV	bf08268	clean	AWS c7i.48xlarge	2026-10-12	`sweeps/results/E-E`
E-E throughput	Section VII-D	n/r	n/r	laptop (WSL2)	2026-10-11	`sweeps/results/E-E-laptop`